A History of the Agricultural Research Center--Hays: The First 100 Years

See http://www.oznet.ksu.edu/pr_histpubs/Pubs/SB663A.pdf for the first half of this publication.

RESEARCH OVERVIEW

No attempt will be made to give detailed or even summary results of the many research projects that have been carried out during the past 100 years. Rather I will endeavor to highlight some general research areas and describe detailed results in only a few cases. More details about research can be found in the special publication An Annotated Chronology of Research Highlights, Agricultural Research Center—Hays. In addition, careful study of the publications listed in Appendix C will reveal some interesting developments. During the early part of the Station history, major emphasis was on demonstrations and production research that could be used immediately by the general farming community. Although that aim is still important and the need is still being met through publications of the Experiment Station or Extension Service and popular outlets, the number of articles published in scientific journals has increased greatly. For example: Prior to 1950, approximately 25 articles had been subjected to peer review and printed in scientific publications. One staff scientist, Mr. Arthur F. Swanson, authored more than half of those. In only about 12 years following 1950, the number more than doubled. That trend has continued and intensified in the ensuing years. The need to provide published information that will enable Kansas farmers to increase their total production and production efficiencies cannot be over emphasized. But the significance of the large number of contributions to the worldwide scientific literature must not be overlooked.

The role of the U.S. Department of Agriculture (USDA) in research at Hays should be recognized. As has been pointed out, much of the early crops research was funded at least partially by the USDA. Research agronomists in cereal crops, forage crops, soil management, weed control, and soil conservation were federal employees. The Soil Conservation Service, USDA, supported studies of soil erosion and water runoff for only a few years. The department began reducing support for forage research in the 1940's and terminated that federal project about 1950. Next, in 1959 the Division of Dryland Agriculture, USDA, terminated most of the funding for the project started in 1906. In 1969, the Agricultural Research Service (ARS), USDA ended funding for cereal crops work, which also had begun early in the century. Finally, in 1973, ARS closed the last federally funded project, weed control. In every instance, the research was continued with support from state funds, but many people felt that USDA officials were short sighted. Because significant state financial support had augmented federal funds, it is doubtful that USDA could have achieved so many meaningful results elsewhere at so small a cost.

Many of the researchers who have worked at the Station are mentioned in the following sections. A complete list is included on page 40.



Crowd assembled for the second annual Roundup, April 1, 1915.

BEEF CATTLE

Although several classes of livestock appeared on the Station immediately after its establishment, beef cattle soon emerged as the most important as far as research and demonstrations were concerned. The herd management and feeding work started in the second year of the Station's operation and has continued without interruption to the present. The first beef herd consisted of 144 head of common cattle, mostly breeding stock. These cattle were sold in 1907, and an experiment was started to compare four beef breeds: Shorthorn, Angus, Hereford, and Galloway. The plan was to maintain these breeds as separate entities, produce calves, and compare the performance of the different breeds. Unfortunately, the experiment was never designed adequately, and little or no usable information was forthcoming. In the following years, the Hereford herd was increased, but little cow herd research was accomplished. Soon after his arrival in 1921, Louis C. Aicher began a systematic dispersal of all the cows except Herefords. He initiated careful selection and breeding policies and developed one of the superior nonregistered Hereford herds in the country. Bulletin 453 describes his program in detail, including the pedigrees of the outstanding sires he used. Little research was conducted with the cow herd, but it did serve as an excellent demonstration of what could be accomplished with good breeding and selection. Calves produced by the cow herd were used in feeding trials, but only after replacement heifers for the cow herd were selected.

Results of the beef cattle work have been reported to the agricultural community through an ongoing series of publications that began with a mimeographed circular in 1913. The first recorded reference to Roundup appears in the 1914 Circular entitled "Roundup on The Fort Hays Reservation." Since that field day on May 1, 1914, Roundup has been a fixture at the Experiment Station. It has been held each year except 1945 when German prisoners of war were housed in the livestock research area. Roundup reports have been published in all but two years, making it one of the longest series published by any public research agency. For many years, the Roundup reports consisted almost entirely of data recorded from the various experiments with little or no interpretation. There seemed to be a reluctance to summarize the data and to formulate any type of recommendation. Rather, the philosophy seemed to be, "Only the facts are given. Draw your own conclusions." This same philosophy was evident in publications from other research projects. Changes needed to overcome



"Entertainment" at the second annual roundup, April 1, 1915.

this reluctance to interpret results were slow to emerge, but surely usable recommendations should be part of publications written to provide information to farmers.

From about 1920 to 1944, Dr. C.W. McCampbell, animal scientist located in Manhattan, was the principal investigator for beef cattle; although Mr. Aicher and his herdsmen supervised the actual day-to-day activities. Mr. Frank Kessler arrived in 1946 and became involved in the beef cattle feeding work as well as some of the range management studies, but he had little input into cow herd management. He remained until 1957, when Mr. John R. Brethour came to the Station. Except for a two year period in the 1960's, he has directed the beef cattle research since that time. He was the first on-site individual to have full responsibility for the project. Brethour expanded the cow herd research to include performance testing, creep feeding, breed evaluations, management strategies, and reproductive physiology. The recent purchase of additional rangeland has allowed for expansion from about 120 to 350 cows. However, this number represents less

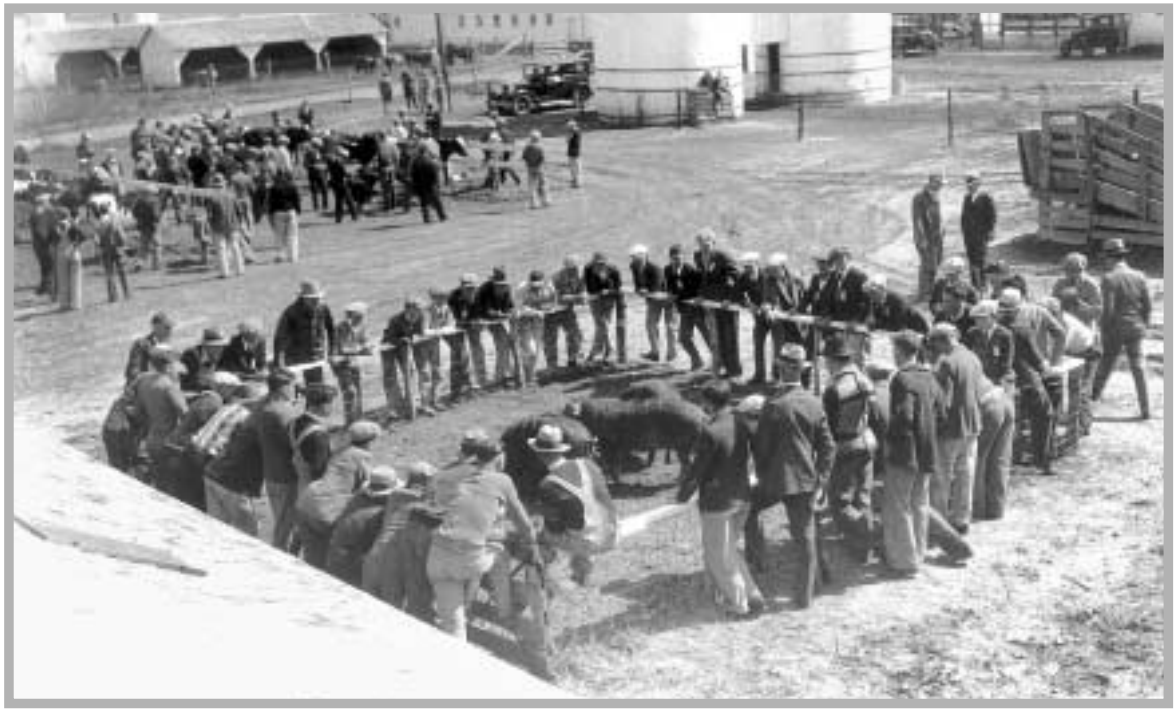


John R. Brethour

than one-fourth of the more than 1,500 head of cattle on the Station. Beef cattle nutrition work always has focused on crops grown in the area. Sorghum has been especially important, and the Station certainly has been a leader in research with this crop. From beginnings early in the century, the studies involved both grain and forage sorghums. The forage research has

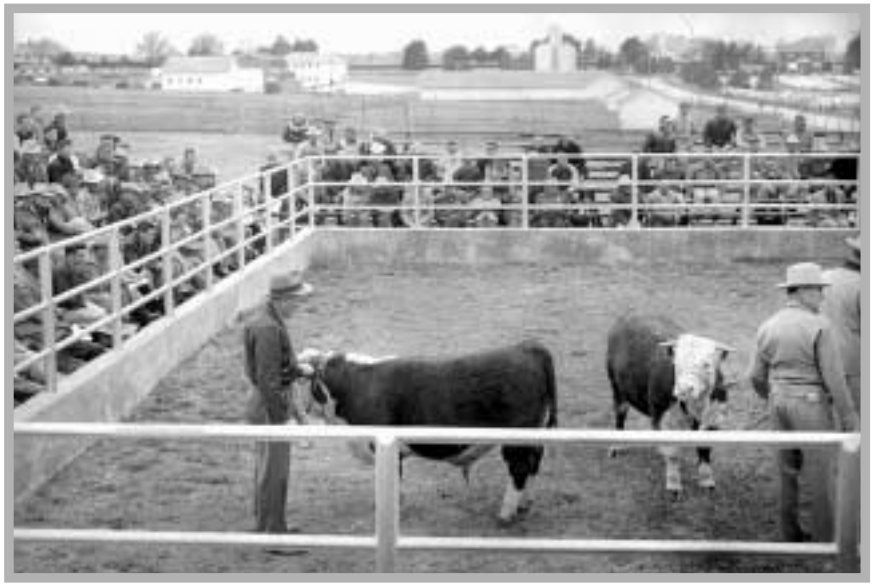


Cars and Crowd at Roundup, c.1926.



Facilities used for Judging Contest, 1932.

shown important relationships between dry matter content and resulting silage quality, demonstrated variation among sorghum varieties, and established that silage additives then available did not improve quality sufficiently to provide an economic response. The Station was among the first to evaluate ensiling high moisture sorghum grain. When fall weather is cool and moist, harvesting dry sorghum grain is often difficult, but ensiling at the proper high moisture content can salvage the grain. Many other silage and dry roughage experiments have been carried out and results presented in the Roundup reports.



Cattle in the improved arena, 1966, but still no shelter from inclement weather.

Research with feed grains directed by Brethour proved that with proper preparation, the feeding value of sorghum grain is approximately 93 percent that of corn. This is considerably higher than the previously recognized conversion factor. John also did pioneering research using wheat in cattle fat-

tening rations. Some preliminary work had been conducted, but had been based on the assumption that wheat could constitute only a relatively small percentage of the total grain in a feeding ration, because higher amounts surely would cause digestive problems. His work showed that, with correct management, feeding wheat as 100 percent of the grain in finishing rations was possible. He also studied different wheat types and varieties and conclusively proved that the commonly grown hard red winter wheat varieties were fully equal to the so-called feed varieties. Probably the most important result of the wheat-feeding experiments was to inform producers that wheat could be used successfully in cattle rations. Estimates indicate that over a half billion bushels of wheat have been fed to cattle in the U.S. since Brethour's work proved it could be used safely. In addition to the conventional feeding studies, many experiments involved feed additives and performance-enhancing implants.

Brethour's extensive use of computers as an aid to conducting and interpreting research results deserves special mention. In the early years of computer science, no software was available to provide the types of analysis that we now have come to expect. Brethour, through self education, wrote programs that met the needs of his research work. He developed an interactive least-cost feed ration program that enabled cattle feeders to input various feed ingredients along with their cost and determine which was the most economical to use in their feedlot. As future beef producers look back on Brethour's many accomplishments, perhaps the greatest will be his work to establish ultrasound technology as a production tool. He was one of the first to recognize the potential of ultrasound readings as an aid in predicting carcass composition prior to slaughter. Without trying to explain the technology in detail, suffice to say that by interfacing the ultrasound system with computers, feedlot managers can scan large numbers of cattle in a comparatively short time and accurately predict the optimum time when they should be marketed. Later, Brethour worked with ultrasound scanning on very young cattle. Results indicate that scanning at weaning time can predict carcass quality when the animal is slaughtered many months later. This ultrasound-computer technology has been assigned to Kansas State University Research Foundation and patents have been granted. The Station is now acclaimed as the world leader in developing ultrasound technology as a beef-cattle production tool.



Enclosed arena with permanent seats for about 800. These facilities were designed by Duitsman.

CEREAL CROPS

Investigations with Cereal crops began with demonstration plantings in 1902. The Station Superintendent conducted, or at least directed, the work. Even at that early date, records indicate that the USDA supplied some funds and furnished some seed stocks. The earliest record of a staff member other than the Superintendent being in charge of the cereal investigations was Mr. A. D. Colliver in 1905. That individual was paid partially from USDA funds, but from then until 1912, the state supplied some money for the project leader's salary as well as furnishing physical facilities and labor. A more formalized arrangement with the USDA began in 1912, and a full time USDA scientist, Mr. F. A. Kiene, was put in charge and remained until 1919. Mr. Arthur F. Swanson, who led the project until he retired in 1951, replaced him.

From 1902 through 1920, most of the cereal investigations involved variety testing along with observations and research concerning rate and date of seeding. Winter wheat received the greatest attention; but other crops, including spring wheat, grain sorghum, oats, barley, and corn were tested. Forage sorghum investigations were carried out by the forage crops project until 1936. After that date and until 1953, all of the sorghum and wheat work fell under one project leader. The early wheat work was done almost entirely with Turkey and Kharkov and selections from those two varieties. Kanred, the first important variety developed at Kansas State College, was grown widely throughout the state. The Station actively participated in its development and was responsible for distributing many thousands of bushels of seed. Later, the varieties Tenmarq and Comanche were developed at the College; they also became important in Kansas and the surrounding areas. Swanson started the active wheat breeding program at the Station. Kiowa was the first variety developed and distributed to farmers. Bison, a close relative of Kiowa, was released in 1955 and soon became the most popular variety of the state. In 1961, Bison was grown on 27 percent of the Kansas wheat acreage and was regarded highly for its milling and baking qualities.

In 1947, a state financed position was added to assist the USDA scientist. Several young researchers filled that position prior to the arrival of Dr. William M. Ross in 1951. He came to Hays expecting to gain valuable training by working under Swanson. However, Swanson soon departed for an international assignment in Peru and retired after his return. Ross found himself in charge, accepted the USDA appointment, and quickly established a reputation as a highly competent plant breeder. In 1953 the wheat and sorghum breeding pro-



Combine mounted on Caterpillar tractor for harvesting experimental grain plots. This machine, designed by Aicher, was no doubt the first self-propelled plot combine.

grams were separated. The USDA employee, Ross, was put in charge of sorghum breeding, and the state assumed responsibility for the wheat breeding work, led by Dr. John D. Miller. The federal sorghum project at Hays was terminated in 1969, but not before Ross was recognized as a premier sorghum geneticist. Since 1969 both wheat and sorghum research has been funded from state funds supplemented by various grants.

During the late 1950's, the concept of hybrid wheat was studied intensively, but conventional plant breeding that led to new wheat varieties was not particularly productive. That changed in 1962, when Dr. Ronald W. Livers assumed leadership of the wheat-improvement project and began actively developing lines and varieties. During his tenure, five important varieties were developed and released from the Station: Eagle, Sage, Kirwin, Larned, and Cheney. For many years Eagle's milling and baking performance set the standard for desirable quality. Larned, released in 1976, is still grown on limited acres in the state. Following Liver's death in 1979, Dr. T. Joe Martin, who was working on the Station as a Plant Pathologist, transferred to the wheat breeding position. Martin carried on the work that Livers had in place and expanded it into a truly outstanding program. The first variety released following his appointment as wheat breeder was Arkan. It became one of the leading varieties in Kansas in spite of some controversy regarding its grade classification. But perhaps this controversy led to some needed changes in wheat-grading procedures. Arkan was followed by Dodge, Norkan, and Ike. In 1990 an experimental hard white wheat, KS84HW196, was released to the American White Wheat Producers Association. This variety initiated production of hard white wheat in western Kansas. In 1999, a high-yielding, hard white wheat, Trego, was distributed to the seed producers as a general public release. The wheat-breeding program at Hays began the conversion to the development of hard white wheat in 1987, and by 1999, was devoting about 90% of its resources to the hard whites. The release of Trego and other high- yielding, high-quality white varieties may significantly change Kansas wheat production and provide greater opportunities in international trade.



Seeding wheat breeding plots with a 2-row tractor mounted planter. As late as the 1950's this was done entirely with manpower.

As important as new ready-to-plant varieties may be to agriculture, perhaps even more important are germplasm lines that can be used by plant breeders throughout the world to develop improved varieties of wheat. In addition to varieties released for farm production, the wheat program has developed and released some 18 or more germplasm lines that were made available to all public and private breeding programs. These included some of the original genetic resources used in the early hybrid-wheat program and newly developed lines with improved co-

leoptile length coupled with short stature, and resistance to Hessian fly, wheat streak mosaic virus, wheat curl mite, green bug, and Russian wheat aphid. Developing varieties and germplasm releases with insect and disease resistance requires cooperative efforts among the wheat breeder, entomologists, and plant pathologists. These contributions will be discussed further in the sections on entomology and plant pathology research.

However, as far as farmers are concerned, new varieties provide the most tangible evidence of a successful wheat-breeding program. Certainly, germplasm releases should not overshadow new variety development. The importance of varieties released from the Station can be illustrated by noting that since Kiowa was released in 1950, new varieties developed at the Station have been planted on more than 120 million acres. This is equivalent to total wheat plantings for approximately 10 years in Kansas. Assessing the monetary value of research is often difficult; however, in the case of new wheat varieties, some observations are possible. A new variety is developed and released with the aim of providing some improvement over the existing varieties that it is likely to replace. This may be increased disease and insect resistance, increased drought tolerance, better quality, and other factors. But finally, the success of a variety is measured by its yield and acceptance by growers. Assume that a new variety will yield only one additional bushel per acre (many surpass that) than existing varieties. Then, if over the years, improved varieties have been planted on as many as 120 million acres, increased production over time has been 120 million bushels, or more than one million bushels for each year the Station has been in existence. Similar extrapolations can be used to evaluate other research.

As pointed out, sorghum was included in the demonstration plantings as early as 1902. Most sorghum varieties were tall forage types with comparatively low grain yields. The first important grain variety, Pink Kafir, was selected about 1912 and was an important part of the grain sorghum production for many years. Swanson believed that sorghum improvement was fully as important as the wheat work, and he began selecting and breeding to improve the existing varieties. Wheatland, the first combine-type sorghum to be grown widely, was brought to the Station in 1929 and released in 1931. Westland, a variety developed at the Garden City Experiment Station, and Midland, developed by Swanson, replaced Wheatland. These were the leading varieties until sorghum hybrids became available in the 1950's. Midland seed sales played an important role in the Station finances during the 1940's. In 1945, approximately 250,000 pounds of seed were distributed. Other grain sorghum varieties developed by Swanson included Early Kalo, Club Kafir, and Cody. He also began extensive work to improve forage varieties. The available varieties, such as Red



A modern self-propelled plot combine.

Amber and Black Amber, were late maturing, often lodged badly, and produced little grain. Early Sumac was developed at the Station prior to Swanson's arrival and largely replaced those older forage varieties. Early Sumac remained important but was supplemented by varieties developed at the Station. Swanson's active sorghum-improvement project led to the release of several forage varieties including Norkan and Ellis. Both had excellent feed qualities, and Norkan had high grain yields, but neither competed well with varieties that had higher forage yields. As with grain sorghum, hybrids soon replaced most of the forage varieties.

Discovery of male-sterile plants (a trait necessary to produce hybrids in crops that may self-pollinate) enabled sorghum breeders to produce hybrids for commercial use. Pure-line variety development essentially ceased. The Kansas Agricultural Experiment Station released some grain sorghum hybrids. However, it soon became apparent the most important function for the Station would be to develop breeding lines that could be used by public and private breeding programs. In 1969, the USDA terminated the sorghum-breeding project at Hays. Ross was transferred to Lincoln, Nebraska, and Mr. Harold Hackerott, who was already on the Station staff, became the sorghum breeder. Research by the sorghum geneticist, working cooperatively with entomologists and plant pathologists, certainly has served a most important role in sorghum improvement. In 1969, a new greenbug biotype that attacked sorghum throughout its production areas was identified. Through intensive efforts Hackerott, and Tom Harvey, Entomologist, found resistance and developed the first germplasm lines needed to combat this new sorghum pest. Since that time, several additional releases have provided resistance to new greenbug biotypes. Lines resistant to several diseases also were developed. Much of the genetic background of many of the commercially grown hybrid sorghums can be traced to breeding lines released from the Station.

Following Hackerott's death in 1986, Dr. Kenneth D. Kofoid was hired to head the sorghum-breeding project. He had been a wheat geneticist at North Dakota State University prior to coming to Hays, but he was well aware of work being done at the Station. Kofoid was Bill Ross's first graduate research assistant after Ross moved to Lincoln, Nebraska. As such, Kofoid had heard much about working at the Hays Station, especially about the weather. While in the field on hot, blustery days, Ross quite often would stand and look around and say, "Ah, just like western Kansas!" Kofoid has continued the close working relationship between the breeding project and both the entomology and pathology projects. When a new biotype of greenbug was identified in 1990, he and Harvey were able to release a resistant germplasm within months. The cooperative effort with the pathology project has shown that sorghum also can be infected with wheat streak



Joe Martin, Wheat Breeder, discussing wheat variety performance plots at a Spring Field Day.

mosaic virus. Although this is not good news to sorghum producers, having this knowledge allows the breeding program to develop germplasm that tolerates the disease. In all, this team has released 27 germplasm lines with insect and disease resistance to the commercial seed industry for their use in developing hybrids.



Kenneth Kofoid

In addition to the work with wheat and sorghum, considerable selection and development work with corn was done early in the century, Climatic conditions in the Hays area were not conducive to extensive corn production using the open-pollinated varieties then available. An exception was a selection that proved to be quite well adapted to conditions in west central Kansas. This variety, later named Hays Golden, was grown widely in Kansas until the advent of corn hybrids and higher-yielding grain sorghums. Hays Golden was used as a parent variety in several commercial corn hybrids. In recent years, corn has again increased in popularity and is being grown on nonirrigated land. Some of the newer hybrids are better adapted to the area, and no doubt corn will continue to be important, at least until prolonged drought occurs.

Other cereals, including spring wheat, winter and spring barleys, and oats were tested and observed in demonstration plantings, but only limited variety-improvement work was done. Briefly, the studies showed that none of those crops could compete successfully with winter wheat and sorghum for grain production in the west central Great Plains. Some spring wheat varieties and germplasms continue to provide genetic material that can be incorporated into the winter wheat- breeding program.

FORAGE CROPS

Experimental work with forage crops began with the establishment of the Station. The most extensive work during those early times was with forage sorghum. After his arrival in 1919, Swanson of the cereal project, did some breeding and selection, but much of the genetic improvement took place after the forage sorghum program was transferred to Cereal Crops in 1936. In addition to observing varieties, the early work included methods of culture and utilization of the crop. Sudan grass received considerable attention; particularly important were cultural practices involving rate and date of seeding and determination of optimum growth stage for harvesting hay. Some efforts were made to improve alfalfa, bromegrass, and big bluestem. Early observational plantings included a large number of different crops, some of which (e.g., soybeans) could not be classified as forages.

Prior to 1913, the Station Superintendent oversaw most of the forage work. In that year a cooperative agreement was initiated with the Bureau of Plant Industry, USDA. The department paid the salary of a scientific worker stationed at Hays, and the Station provided the physical facilities and most of the labor. This was similar to the arrangement with other USDA scientists located at Hays. This cooperative venture continued until 1937, when the state assumed responsibility for one-half of the scientific worker's salary. All federal support for the project was terminated in 1950. Mr. R.E. Getty, the first USDA scientist, arrived in 1913 and directed the work until 1929. He tested a large number of forage and miscellaneous crops to determine their adaptability to conditions in west central Kansas. His tests included several legumes in addition to soybeans, and he observed some root crops, e.g. sugar beets. Although Mr. Getty used the best varieties then available, he soon determined that few, if any, of these alternative crops could compete successfully with wheat and sorghum under the climatic conditions at Hays. Farmers had considerable interest in foxtail millets. They could

be grown successfully, but tests showed that they could not compete with forage sorghum in either yield or forage quality.

During the drought of 1932-1936, survival of native grasses was studied carefully under the direction of Mr. David A. Savage. In anticipation of a need to reseed after the drought, studies were initiated to determine optimum seeding date, seedbed preparation, and seeding methods. As will be seen, these studies were expanded in later years. The breeding program, or perhaps more accurately a selection program, for improving buffalograss began in 1936. A selection that was superior in seed production and height of seed stems was located and increased vegetatively in 1942. This female plant selection then was pollinated with numerous selected male plants. This improved selection was released as a buffalograss variety named Hays 1-I, and several thousand pounds of seed were harvested and distributed. Concurrent with this buffalograss variety improvement work was the research needed to develop a system for improving the germination of newly harvested buffalograss seed. This process is discussed in some detail in the section on the L.C. Aicher administration. Most of the grass research work in the forage project was terminated with the arrival of Dr. John Launchbaugh in 1955 as leader of the Range Management project. As noted in the Range Management section, he did outstanding work in range reseeding and other aspects of grass management.

It was apparent as early as 1902 that alfalfa could be grown successfully on the bottomland fields along Big Creek. Problems associated with establishment and seed production were studied. Later, lines and varieties, including the bacterial wilt-resistant variety Buffalo, were evaluated. An active alfalfa breeding project was initiated in 1954 when the spotted alfalfa aphid first appeared. The plant breeder, Harold Hackerott, and the entomologist, Tom Harvey, worked together to find resistance to this insect pest, and, certainly in nearly record time, developed and released the resistant variety Cody. A similar program a few years later resulted in the release of Kanza, the first alfalfa variety resistant to the pea aphid. This cooperation between scientists from different disciplines illustrates the value of having close association and cooperation among the resident staff members.

Hackerott transferred to the sorghum breeding position in 1969, and only limited work was accomplished on cultivated forages other than sorghum after that. As noted above, the Range Management scientist did most of the grass research. A new project was started in 1971 to investigate some possible alternative crops for use in west central Kansas.

ALTERNATIVE CROPS

Hard red winter wheat became the crop of choice soon after it was introduced into the area late in the 19th century. However, farmers then and continuing to the present have sought possible alternative crops that would equal or exceed returns from wheat. The first of these crops that became important, of course, was sorghum; but during the early years of the 20th century, considerable acreage of corn also was planted. Since the Station's origin, many other possible alternative crops have been planted and observed by crop scientists. As noted previously, most of these crops were discarded quickly as unsatisfactory for the climatic conditions or simply did not have potential for economic returns that could compete with those of wheat and sorghum. This was true even with corn, especially following development of combine-type grain sorghums. More recently, with improved corn hybrids and soybean varieties, these two crops have extended their area of adaptability further west in Kansas than previously thought possible.

Research to find alternative crops intensified with the arrival of Mr. William D. Stegmeier in 1971. He had worked previously at the Garden City Branch Experiment Station, where his studies with wheat, sorghum, corn, and several possible alternative minor crops

such as castor beans, soybeans, pinto beans, safflower, winter barley, and popcorn provided an excellent background for his studies at Hays. When he came to the Station, most of the oil-type sunflower production was in Texas and in the northern states. Stegmeier believed that the crop had potential in Kansas and began an intensive breeding program to improve yields and increase disease and insect resistance in sunflower varieties. His work contributed to acceptance of the crop by Kansas farmers. Acreage was limited for several years, partly because seed-processing plants were not available within the state. With the addition of such plants, sunflower is now an established crop in the state.

Stegmeier's most notable work was his breeding program with pearl millet. This crop, which is entirely different from the previously tested foxtail and proso millets, is a staple food and feed crop in many countries of Asia and Africa, but it had never been grown in Kansas. He believed that it had potential as a new crop for Kansas farmers and that it might compete successfully with grain sorghum. When he started his work, the typical pearl millet variety was a tall late-maturing type adapted to tropical climates. Stegmeier knew that, if the crop was to be successful as an alternative to sorghum, it must be converted to a short-stature, early-maturing, high-yielding grain type. His breeding work soon attracted attention beyond the borders of Kansas. Beginning in 1978, the project at Hays was linked to international crop improvement programs including the International Crops Institute for Semi-Arid Tropics (ICRISAT) headquartered in Hyderabad, India, and the U. S. Department of State Aid to International Development program entitled International Sorghum and Millet Cooperative Research (INTSORMIL). That cooperation resulted in exchange of several thousand varieties, germplasm lines, and elite inbred selections of pearl millet.



William Stegmeier

Developing hybrids of crops such as sorghum and pearl millet, requires parental lines that are male sterile (that is, plants that will not self pollinate), thus assuring fertilization from a separate male line. Stegmeier's breeding program at Hays developed two male sterile lines from a West Africa source. These lines have been highly successful as parental stock and were recommended by the Indian National Program for general use as seed parents. The new inbred lines imparted early maturity, larger seed size, and higher grain yields than had been available previously. Early maturity is particularly important in areas of India and Africa, where short season crops are desirable because of the need to produce two or even three crops during a given year. Reliable reports from India indicate that by 1997, at least half of all of the pearl millet hybrids marketed by India's private growers were produced using these female parents. This means that more than five million acres of pearl millet grown in India in a given year are hybrids derived from Stegmeier's program. Perhaps even more importantly, these inbred lines are used widely in breeding programs in India and elsewhere. This will lead, in turn, to further improvement in this significant feed and food crop. Because of his work with the international sector, Stegmeier was recog-

nized as part of a team that received a distinguished scientific achievement award sponsored by ICRISAT. Pearl millet is not yet grown widely in Kansas and perhaps never will be, even though extensive yield trials showed that the new hybrids can compete successfully with grain sorghum. Feeding trials with beef cattle indicated that the grain and forage could be used in place of sorghum grain and forage. But the impact of the research program at Hays has been felt worldwide and is particularly meaningful in areas that are often short of food to feed their population.

In addition to the research with pearl millet and sunflower, Stegmeier did a limited amount of work with canola, amaranth, and other possible alternative crops. Of these, probably only canola may find a niche in Kansas agriculture and then only after continued breeding and varietal improvement has taken place. Stegmeier's untimely death from cancer in 1998 brought closure to one of the important crop-improvement programs in the world.

ENTOMOLOGY

Alfalfa flowers must be "tripped" to produce seed. The need to determine the roles of insects and various mechanical devices in alfalfa flower tripping intensified in 1948. Some cooperative work with Agricultural Engineering to study mechanical tripping was underway. The first full-time entomologist arrived at the Station in 1948 to determine the role of both beneficial and harmful insects. This established a unique arrangement between the Station and the Department of Entomology, Kansas State University, in that the entomologist at Hays was a member of the Department staff at Manhattan. This arrangement has continued to the present time and is the only case where a full-time scientist located at Hays is not a Station faculty member.

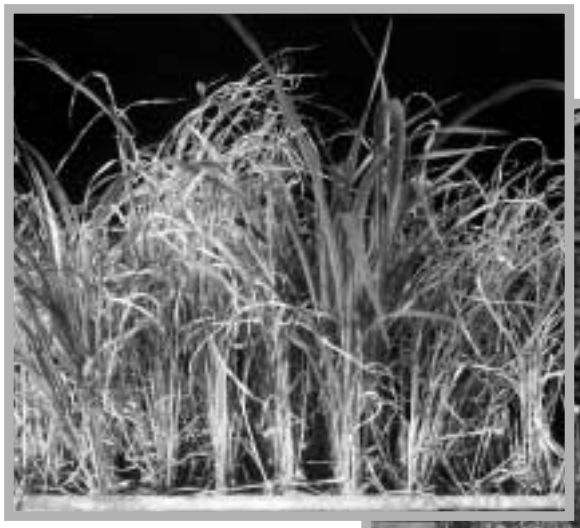
Dr. Woodrow W. Franklin, who had been working on alfalfa insects in Manhattan, filled the position. He immediately began research to determine the role of wild bees as



Reducing grasshopper infestation with a "hopper dozer." Modern insecticides and insect-resistant crops have reduced losses due to insects.

well as colonies of domestic honeybees in the pollination of the crop. He had sufficient colonies to warrant harvesting and processing honey. Additionally, Franklin did extensive survey work throughout west central Kansas to determine levels of insect infestation and species of insects that were affecting alfalfa. He accepted an overseas assignment in 1953 and Mr.Tom L. Harvey replaced him in 1954. Alfalfa pollination research was largely abandoned. Harvey has remained at the Station since that date and epitomizes the definition of a long-term commitment to a research program. A few years after his arrival, he took sabbatical leave to pursue and obtain a Ph.D. from Oklahoma State University.

Although his independent research in several areas has been important, perhaps Harvey's greatest contribution has been to work closely with other staff members to develop insect-resistant crop germplasm lines and varieties. He also worked closely with livestock researchers to develop unique and effective insect control methods, particularly for beef cattle. His impressive list of publications documents his contributions, but one needs special mention. He originated the idea for, and published the first technical paper on, insecticide-impregnated ear tags for fly control on cattle. This system soon became an industry standard. Good researchers are never short of ideas, but some ideas work, whereas others do not. This is illustrated by the studies with hornfly control on cattle, Harvey and John Brethour, Animal Scientist, were trying to develop a bolus (a large pill) containing a larvicide that eventually would be excreted in the manure and contain sufficient insecticide to kill hornfly larvae. They had trouble getting a bolus large enough to contain the necessary amount of larvicide and decided to try using a 6-ounce frozen juice can as a mold. They then tried to administer the bolus down the animal's throat using a caulking gun. Although this bolus was too large for practical application, it proved to be a forerunner of the sustained-release boluses now used to treat cattle. The livestock insect research is documented in 24 articles published in refereed journals from 1959 to 1988 (see appendix C). This work was discontinued in 1989 when an increased effort was needed to study insects and mites in relation to the transmission of virus diseases of wheat and sorghum. That research involves close cooperation with Dallas Seifers, plant pathologist, and one



Dead sorghum plants killed by aphids vs. resistant lines developed through plant breeding and selection.



Tom Harvey

notable result was the identification of the wheat curl mite as the vector of the High Plains virus of wheat and corn.

Several insect-resistant varieties and lines of several crops are mentioned in other research sections. Teamwork between plant breeders and the entomologist resulted in the development and release of such varieties as Cody and Kanza alfalfa. They identified a greenbug-resistant sorghum and developed and released several germplasm lines used in commercial and public breeding programs to produce greenbug-resistant sorghum hybrids. Likewise, they identified, developed, and released wheat lines resistant to the wheat curl mite, the vector for the destructive wheat streak mosaic. More recently, a Russian wheat aphid-resistant wheat variety, Stanton, was released. Prior to doing the extensive research necessary to develop these varieties and lines, it often was necessary to learn to rear the insect in question and then to effectively infest plants being used in the study. Harvey has a unique ability to adapt systems necessary to accomplish these tasks.

PLANT PATHOLOGY

In 1952 a severe outbreak of western wheat streak mosaic virus devastated thousands of acres of the Kansas crop. Thus began many years of research to find resistance to the disease and incorporate that resistance into varieties and lines. In 1952, the USDA funded a position at Hays to study this largely unknown disease. The program was fairly short-lived and produced little or no usable information. From then until 1974, the disease resistance work was done by the wheat breeder, Dr. Ron Livers, in cooperation with pathologists at the central Station in Manhattan. In 1974, the Kansas Wheat Commission provided funds to substantially finance a new pathologist position at the Station. Dr. T. Joe Martin began working closely with Livers in the wheat breeding program. Following Livers' illness and death in 1979, Martin assumed the wheat breeding responsibilities, and Dr. Kurt Bender was employed. Unfortunately, responsibilities required that he return to the family farm in Iowa, so his stay at the Station was quite brief. Dr. Dallas L. Seifers assumed the position of plant pathologist in early 1982 and has remained in that position.



Dallas Seifers

The plant pathologist works closely with plant breeders to develop disease-resistant varieties and lines. The original thrust of the pathology work was to find resistance to wheat streak mosaic virus. In that effort and to further expand the plant pathology work, Seifers delved into more basic studies. That work is made possible in part by continuing support from the Kansas Wheat Commission and the Kansas Crop Improvement Association. The research now includes other viruses that cause losses in wheat, sorghum, and other crops. Several of these infect sorghum, including maize dwarf mosaic virus, sugar cane mosaic virus, Johnsongrass mosaic virus, and sorghum mosaic virus. Additionally, vigilance is required to identify new viruses that may affect crops. Technological advances have permitted expansion into even more basic research areas, including work with transgenic (gene transfer) wheats. In recent years cooperative projects

have begun with scientists in several states, Canada, Israel, and France. Pioneering work at Hays enables the project to supply scientists throughout the United States and other countries with virus isolates and procedures for use in viral research work.

RANGE MANAGEMENT

When the Station was established, only a small part of the land allocation was under cultivation. Additional sod was broken during the early years, but about half of the original acreage remained in native grasses. Little management research was done on these pastures prior to 1946, and grass improvement work was carried out on the Forage Crops project. In 1946, a new position was added, and Mr. Frank B. Kessler was hired. He served a dual capacity in that he began pasture management studies and also was involved in some of the research on beef cattle feeding. At about the same time, Mr. Friedrich E. Meenen was named project leader of the Forage Crops project, and both he and Kessler were involved in range management and grass improvement. Meenen continued ongoing work to select and develop new strains of buffalograss, but with his departure in 1951, most of the forage improvement research emphasized cultivated crops rather than native grasses.

A study started in 1946 compared heavy, moderate, and light intensities of summer grazing on native shortgrass range. These three pastures were maintained under the same grazing regimes for many years, but additional research was superimposed to determine the role of supplemental feeding on grass. In 1948, buffalograss, western wheatgrass, and intermediate wheatgrass were planted separately on three previously cultivated lowland fields. Grazing comparisons on these three grasses were made from 1949 to 1960. Not unexpectedly, the tall-growing cool-season wheatgrasses produced more forage than buffalograss on this lowland site.

Pasture and range research expanded rapidly beginning in 1955 with the arrival of Dr. John L. Launchbaugh. During the next 30 years, he conducted studies on range reseeding, winter grazing on native shortgrass range, use of protein supplements with summer grazing, fertilization of native pastures, and weed control in both native and reseeded pastures. The range reseeding research was no doubt the most extensive ever undertaken. Launchbaugh pioneered work with complementary forages used in conjunction with grazing native rangeland. Briefly, that involved using cropland adjacent to pastures to produce forage that cattle could graze directly from the field. The system significantly improved cattle performance. Launchbaugh initiated yet another line of research with his studies of intensive-early stocking on shortgrass ranges. This work on the shortgrass areas was done essentially concurrently with research at the central Station in Manhattan on the tallgrass prairies in the Flint Hills area. Launchbaugh's publications relating to reseeding research, grazing intensity, and other work in range management in Experiment Station Circulars and Bulletins and in technical journals brought recognition from the ranching and scientific communities. Today, many of his recommendations form the back-



Keith Harmoney

ground of sound range management practices on native shortgrass pastures.

Following Launchbaugh's retirement in 1985, Dr. Kenneth C. Olson was employed to carry on the range research. He was followed in 1993 by Dr. Eric Vanzant. Both of them continued and refined the intensive-early stocking research that had been started by Launchbaugh. They found that shortgrass rangeland does not respond to intensive stocking as positively as Flint Hills range, and unless carefully managed, the system will have some undesirable effects on the forage productivity and the plant composition. In addition to studies relating directly to rangeland, research was initiated with several introduced warm-season grasses and protein supplementation. Since about 1990, the range project has involved livestock management as well as grassland management. Dr. Keith R. Harmoney replaced Vanzant in June of 1999.

SOIL MANAGEMENT

Observations, demonstrations, and a small amount of research relating to soils began as early as 1902. As was the case with most of the research at that time, the Station Superintendent was in charge of the work. But very early in the century, the Division of Dry Land Agriculture, USDA, established a number of experimental stations throughout the Great Plains to study crop adaptation and the effects of different cultural methods, cropping systems, and fertilizer practices on crop production. The project at Hays was started in 1906, and the first scientist, Mr. L.E. Hazen, arrived in 1907. He stayed only until 1908, when Mr. A.L. Hallsted replaced him and served as the project leader until 1945 (36 years). The investigations included many different methods of seedbed preparation; continuous cropping compared to alternate crop-fallow rotations for winter wheat, barley, corn, kafir, milo, oats, and spring wheat; and other crop culture studies. Halstead soon recognized that winter wheat and sorghum would be most important to the area, so over time, most other crops were dropped or became less important in the research. Management studies included uses of commercial fertilizers, barnyard manure, and green manure crops and the effect of burning stubble following grain harvest. Tillage tools used during the early years included the moldboard plow, chisel, subsoilers, and several types of listers. Later, disc-type implements were added, and in the 1940's, the duckfoot and subsurface machines became more important. Over the years, more than 50 cropping systems were studied. Many of these were interwoven with various tillage systems and fertilizer practices. During the 100 years of soils research, hundreds of thousands of soil moisture samples have been taken to determine the effect of treatment on water storage in the soil. For many years, this was one of the labor-intensive tasks on the Station. Driving steel sampling tubes eight or more feet into hard clay or clay-loam soils is laborious. Pulling them out was harder, even though done one foot at a time. Power equipment became available only after about 1950.

The sheer physical size of the project was daunting. This work was done on 1/10th acre plots, each of which measured 2 x 8 rods (33 x 128 feet). When roadways necessary for turning equipment were added, the several hundred plots obviously required a large area of land. As is always the case when conducting soils experiments, having uniform conditions throughout the experimental area was difficult; and the large area required for these tillage studies made for a troubling problem. Some scientists might be critical of the results of these early studies because of this lack of soil uniformity and, at times, lack of replication. However, without a doubt, important results were obtained. The century of investigational work indicates that successful dryland farming in the area is possible, but the farmer must be willing to follow a flexible system when choosing cultural and other management practices.

The cultural studies that started as early as 1907 dominated the soils research well into the 1950's. But beginning about 1950, the general character of the research began to change. Following Mr. Halstead's departure, Mr. Andrew B. Erhart was project leader from 1946 to 1948. Then Mr. Paul L. Brown assumed leadership and, although he did not neglect the ongoing long-term studies, he began some additional work including some more fundamental research. These studies included nitrogen and organic carbon changes in some of the cultivated western Kansas soils. As part of his research in pursuit of his Ph.D. degree, Brown also studied evapotranspiration and water-use efficiency of grain sorghum grown under different cultural practices. Shortly after his departure in 1956, the federal government stopped supporting the lead scientist on the project, and for several years, the ongoing tillage work was maintained, but little new research was undertaken.



Carlyle Thompson with a deep tillage tool used to improve water intake into the soil.

The state assumed total financial responsibility for the soils project and, beginning in 1964, Mr. Carlyle Thompson became the



Applying fertilizer to plots that previously had paper and yard wastes incorporated into the soil.

leader. He also continued many of the ongoing long-term tillage and cultural experiments but began expanding the work to include extensive off-station soil fertility research. During his time as project leader, Carlyle has conducted studies on more than 200 farms in Ellis and surrounding counties. By working away from the main Station, he was able to study the effects of fertilizer on several different soil types with different management practices. Reporting detailed results of all these experiments is not possible here, but several summary statements can be made. Not surprisingly, crop yields increased with increasing depth of moist soil. Also, the judicious use of fertilizer increased water use efficiency. Thompson's work established optimum rates for nitrogen and potassium fertilizers. He studied many different carriers for these fertilizer products as well as distribution and placement systems. More recently, the soils research has included adding waste paper products and yard wastes to agricultural land. Such wastes usually are consigned to landfills.

The wheat-sorghum-fallow crop rotation has been used in west central Kansas for at least 50 years, both under conventional tillage and the newer reduced-tillage systems. Soil management studies have shown that this crop rotation has fewer economic risks and greater yield consistency than wheat-fallow, sorghum-fallow, continuous wheat, or continuous sorghum. This rotation has been adapted widely in the wheat-fallow areas in the western one-third of Kansas. The early work with reduced tillage for crop production was done on the weed research project, but Thompson expanded the research with detailed fertilizer studies and methods of seeding into untilled soil.

WEED CONTROL

Controlling unwanted vegetation is a major problem wherever crops are produced. Often these weeds are introduced as contaminants in crop seeds brought in from other areas. Field bindweed is a fitting example. This deep-rooted perennial weed was introduced into Kansas from Europe, almost certainly in seed of Turkey wheat and perhaps in garden seeds. The weed appeared in many parts of central Kansas at about the time the Station was established. Bindweed competed vigorously with wheat and other crops and caused substantial yield losses. Kansas legislators became aware of the problem, and passed a bill in 1907 to appropriate $1,000 to be expended under the direction of the Board of Regents for experiments to eliminate bindweed. This money was placed with the Department of Agronomy at the College in Manhattan, but because the problem was most acute in central Kansas, the work was undertaken at Hays. However, the weed was not extensive on the Station, so land heavily infested with bindweed was rented from private owners.

The first experiments consisted of 16 treatments with smother crops and cultivation



Planting sorghum in undisturbed wheat stubble, thus the soil had been protected from erosion since the previous year's wheat harvest.

systems. The agronomy professor in charge of the work reported that ordinary methods of tillage during a single season seemed to result in more favorable conditions for bindweed growth. However, winter plowing and the proper use of smother crops such as sorghum and kafir showed some promise to destroy or at least weaken bindweed. He found it could also be destroyed by poisoning the soil with a large quantity of salt or brine, but nothing would grow on the treated soil for several years. The state appropriations were not continued, and the work was discontinued. From then until 1936, various ways to control or eradicate bindweed were tested sporadically, but, except for some sodium chlorate and salt applications, little research was conducted. The conclusions reached in 1908 concerning the use of salt to kill the weed and its effect on soil was certainly prophetic. Applications made in 1919 were still influencing crop production at least into the 1960's.

In 1935 the USDA started a series of bindweed-control studies at research locations throughout the Great Plains. One of those was at Hays, and Mr. F. Leonard Timmons was appointed as project leader. As with other USDA scientists at Hays, USDA paid his salary, and the Station paid most operating costs of the project. This financial arrangement continued until 1951 when the State assumed responsibility for part of the scientist's salary. Then in 1973 all of the federal support was withdrawn, and the State funded the weed control research.

When Timmons arrived, sodium chlorate was the only significant herbicide (other than salt) available. This material did not render the soil sterile for nearly as long as salt, but it did limit crop production for several years and so was adapted for use only on comparatively small areas of the weed. Timmons began cultural control experiments using intensive cultivation and competitive cropping systems. He initiated detailed studies of the root system and its food reserves as affected by frequency and depth of cultivation. Although these root reserve studies did not relate directly to controlling the weed, they led to more complete understanding of the results following various cultivation treatments. He developed practical and effective cultural systems with specific recommendations regarding tillage frequency, competitive crops, and crop rotations. Timmons also initiated studies on the duration of viability of bindweed seed in the soil. His work, along with some that was carried on after his departure, indicated that the seed may remain viable in the soil for more than 50 years.



Sorghum growing in the previous year's wheat stubble. The Station began using this system in the 1960's.

Although field bindweed control was Timmons' primary concern, he expanded weed control work into other areas. The greatest change came in 1945 when the new herbicide 2,4-D became available. Timmons was one of the first civilians to be allowed to test this herbicide, which had been developed for possible military use during World War II (but was not used). It was the first highly selective, growth-regulating

herbicide and marked the beginning of expanding interest in weed control research. Timmons initiated a large number of experiments with 2,4-D for control of bindweed and common weeds in wheat and other Kansas crops. His and subsequent work showed that 2,4-D would not completely eliminate bindweed under most conditions. But it did provide a high degree of control, particularly when combined with some intensive cultivation and competitive cropping systems. Timmons left Hays in 1948 and was succeeded by Mr. William M. Phillips.

Shortly after Phillips became project leader, several objectives were articulated more specifically. Field bindweed research remained important, and Phillips' publications on its control became authoritative in the farming and scientific communities. But as many new herbicides were developed, the research became more wide ranging. Under his direction, the first of many preemergence-type herbicides were evaluated for controlling weeds in sorghum. Expanded use of 2,4-D in wheat led to questions concerning possible carryover of the chemical into the grain. In the mid 1960s Phillips collaborated with scientists in the USDA and other federal and state agencies to conduct comprehensive studies on the effect and fate of 2,4-D applied to wheat. This research showed that 2,4-D applied to the growing crop did not change the milling and baking quality of wheat. These studies contributed to knowledge necessary to establish safe and effective directions for use. Phillips continued searching for ways to control field bindweed and did extensive research on soil persistence and movement of several herbicides. He and Range Management project leader John Launchbaugh studied the root system of the widespread annual weed kochia. Their publication stands as a definitive treatise on the depth and lateral spread of kochia roots.

Phillips did pioneering work to develop a reduced-tillage system for use in wheat-sorghum-fallow rotations. His recommendations enabled producers to combine herbicides with minimum tillage for weed control during fallow after wheat and prior to planting sorghum. His publications in the popular press and professional journals led to acceptance by farmers and recognition by weed control scientists in other states and nations. The system is used widely in many wheat-sorghum (or corn) producing areas. The original recommendations have been altered based on additional research, but the concept remains unchanged. This system decreases the number of field operations and often substantially increases sorghum yields. After its introduction, the herbicide glyphosate (Roundup) quickly became important in reduced-tillage work and other weed-control situations. Phillips was one of the first to observe and study the effects of water quality, spray volumes, and herbicide mixtures on the efficacy of glyphosate.

Prior to the advent of computers, calculating amounts of herbicides and other agricultural chemicals to be applied on small experimental plots to give specific rates per acre was time- consuming and subject to frequent errors. In 1962, Phillips designed a logarithmic-based, slide-rule calculator that quickly



Phillip Stahlman

and accurately determined the amounts needed of both liquid and dry formulations of any percent active ingredient. Following his publication, one of the major agricultural chemical companies refined the original design and furnished the device to many scientists in several disciplines.

Subsequent to Phillips appointment as Acting Station Superintendent in 1976, Mr. Phillip W. Stahlman was hired to continue the weed control research. Along with his many other studies, he continued the work started by Phillips to further determine the effect of several factors on activity of glyphosate. Stahlman conducted extensive studies for control of grassy weeds in wheat, including determining infestation levels that justify applying herbicides for downy brome control. He has been instrumental in developing information on use, rate, timing, and herbicide carryover necessary for labeling a number of new herbicides for weed control in wheat. While continuing his research program, Stahlman furthered his education and obtained a Ph.D. from the University of Wyoming. He has greatly increased the financial support of the project by receiving grants from several sources. Through publications and public appearances, he has become a recognized authority on weed control in the fallow, winter wheat, and grain sorghum producing areas and has national and international recognition.

One unique area of research investigated the use of indigenous soil bacteria for selective control of winter annual grass weeds in winter wheat. Several hundred bacterial isolates were screened for growth-suppressive effects, and nearly 200 were found that inhibited root growth of winter annual brome species and/or jointed goatgrass without affecting winter wheat growth. Some isolates provided up to 70% early-season suppression of downy brome in field experiments. A U.S. patent was granted for three isolates. But, unfortunately, the purified isolates became less effective each time they were cultured. Either they mutated and lost the ability to produce growth-suppressive toxins, or had no need to produce the toxin when they were taken out of the wild. What initially appeared so promising ended in failure, and the project was terminated. However, the potential of using rhizobacteria as biological control agents was demonstrated, and federal scientists are continuing work in this area.

OTHER RESEARCH AND OBSERVATIONS

Although some of the work discussed here certainly qualifies as research, other activities were primarily demonstrations to help area farmers in their quest toward profitable agriculture. These brief descriptions summarize only some of the important information generated outside the major research projects.

GRAIN STORAGE: Combine harvesters came into widespread use in the 1920's and introduced an entirely new grain-handling system. Farmers always had bound wheat when the grain was too moist for safe storage. The ripening process was completed in shocks or stacks. The combine changed that, but no definitive information was available concerning the safe moisture content of stored wheat. During the latter part of the 1920's and the 1930's, high moisture content probably damaged 10 to 25 percent of the total Kansas wheat crop.

Work was started at the Station in 1929 under the direction of Prof. F.C. Fenton, Department of Agricultural Engineering, and Dr. C.O. Swanson, Department of Milling Industry, Kansas State College. They stored wheat in several types of bins with several ventilation systems. Further research was undertaken in 1936 in cooperation with the Bureau of Agricultural Chemistry and Engineering, USDA. Their conclusions can be summarized in a few general statements. Wheat could be stored safely for several years in bins, if moisture content is 13 percent or less. But for wheat with moisture contents in the range of 13 to 14.5 percent, some ventilation appeared to be necessary. Some bin construction types provided sufficient natural ventilation. In other cases, the ventilation could be provided by power fans. Finally, wheat with moisture above 14.5 percent, and perhaps as high as 18 percent, could be stored in ventilated bins, if the distances air traveled were very short and the blowers had high capacity.



Mr. Aicher recognized the value of combine harvesters as early as 1927, but farmers often harvested high moisture grain.

IRRIGATION: Research with nonirrigated agricultural production always has been the main thrust of the Station's work; however, some irrigation investigations were conducted as early as 1903. Like much of the early crops work, this was in cooperation with the USDA. It quickly became apparent that neither underground water sources nor stream flow in the Hays area were sufficient for large-scale agricultural irrigation, so actual irrigation research work was very limited. However supplemental irrigation has proved important in providing water to save valuable plant-breeding material and to produce thousands of tree seedlings grown and distributed from the State Forest Nursery.

SOIL CONSERVATION: This work was started in the 1930's and continued into the early 1940's. It was done in cooperation with the Soil Conservation Service (SCS), Bureau of Chemistry and Soils, USDA. For a few years the SCS provided research scientists, but Superintendent Louis C. Aicher oversaw the work in later years. The studies were designed to determine: (1) loss of soil and water from contoured farm fields and pastures, (2) effects of contouring on the production and succession of native grasses under grazing conditions, (3) methods of establishing native grasses, and (4) the best grasses to use on terrace outlets and flood-water courses. A rather elaborate system of plots and water collection flumes was constructed to measure the water, soil, and nutrient losses from the various plots each time rain resulted in runoff.

HORSES AND MULES: The Station depended chiefly on horses and mules as power for the field work well into the 1920's. The first work animals were purchased in 1903. A few years later, someone in administration suggested that the Station engage in a breeding program that would supply the necessary work stock. Several mares were purchased, and a stallion and a jack obtained. This effort proved to be only moderately successful, but perhaps demonstrated that, although some replacement stock could be produced, most individual farmers could not successfully raise all of the needed horses and mules.

DAIRY CATTLE: The first cow owned by the Station was purchased to supply milk to the Station personnel. Then in 1911, the state legislature appropriated $3,000 to start a dairy project. The intention was to establish the dairy farm as a separate and distinct unit from the rest of the Experiment Station and to operate it in a practical businesslike manner in order to demonstrate the desirability of more dairying for western Kansas. Apparently the dairy herd was never meant to be involved in meaningful research. The dairy was maintained until 1934, when severe drought resulted in poor pasture conditions and limited feed supply, thus forcing the Station to reduce its livestock population. The dairy herd was leased to some state agencies in eastern Kansas. At the termination of the lease agreements, the agencies purchased the animals. The dairy never was reinitiated, so the Station has not maintained a dairy herd since 1935.

SWINE: The first hogs were purchased in 1904. Three breeds were obtained, and feeding experiments were started in 1905. Much of the interest was centered on utilizing hogs to salvage feed that passed through cattle undigested. Other trials compared rations built around crops commonly grown in the area. A few years later, the swine herd was reduced to only one breed, Duroc Jersey. This herd was improved through purchases, breeding, and selection, and by 1928, only purebred Duroc Jersey hogs were on hand. This herd represented some of the best breeding in the state, and many gilts and young boars were sold to farmers throughout western Kansas and eastern Colorado. But few reports of research findings were published. In 1929, 42 sows farrowed and produced 373 pigs of which 299 were saved. Later, intestinal parasites began causing trouble, and some work was carried out to determine how they might be controlled. High temperatures and the extreme drought in the 1930's resulted in death losses, and the Duroc Jersey herd was dispersed. Without apparent

explanation (were they believed to be more heat tolerant?) a herd of Poland China hogs was started, but it never achieved large numbers. Between 1940 and 1950, the only use made of swine was to utilize waste grain. Because the project was not contributing to the experimental findings of the Station, a decision was made to discontinue swine production in 1950. Swine again made a brief appearance on the Station later in the 1950's in an effort to utilize the feedlot waste, but little or no research was accomplished, and the herd again was dispersed.

SHEEP: The first and only sheep owned by the Station were purchased in 1912, and by 1915, the flock consisted of 90 ewes bred for spring lambing. The sheep work had two stated objectives: (1) to determine the possibility of exterminating bindweed with sheep and (2) to demonstrate to local farmers that sheep production could be profitable in west central Kansas. The sheep grazed on bindweed-infested areas until they had eaten the weed down to very short stubble. However, it failed to supply sufficient forage to sustain the animals, and the grazing did not exterminate bindweed. Losses from wild dogs and other factors made it advisable to discontinue the sheep project in 1922. No sheep have been kept on the Station since that time.

FOREST NURSERY

Nursery plantings were made as early as 1903 and included about 1,000 deciduous trees; 4,000 cedar and pine trees; 300 fruit trees, and some 500 vines. In 1905, more nursery plantings were made and approximately 2,500 trees were transplanted from the nursery to the State Park. This was the first of many plantings made by the Station in the park. In 1907, extensive plantings were started to demonstrate the value of windbreaks, particularly those that might protect orchards. Drought and later a hailstorm destroyed many of those trees. A more formalized program started in 1910, when a State Forester was appointed. He established a nursery to produce forest and shade trees, ornamental trees and shrubs, and vines that might grow in the area. In 1912, the Board of Regents directed that the nursery stock be grown at the Station and made available to Kansas residents at cost. Propagation and distribution of nursery stock continued for several years with little or no emphasis on research. This changed in 1927, when Mr. E.W. Johnson, a graduate in Horticulture and Forestry from Colorado Agricultural College, was appointed State Forester. Mr. E.P. Eshbaugh, Mr. J.C. Crupper, Jr., and Mr. J.G. Harrison followed him. All were competent nurserymen and in addition to producing nursery stock for sale, they did some research with new trees and shrubs that might be adapted to western Kansas conditions. Distribution of forest tree seedlings in cooperation with the U.S. Forest Service as authorized by the Clarke-McNary Act was started in 1928. About 10,000 seedlings were distributed in the first year. That number reached as high as 868,000 in 1948. In the 10 years prior to the program being discontinued in 1952, more than 5,000,000 trees were distributed.

In addition to the forest nursery work, the project leader in forest investigations was responsible for orchards and vineyards on the Station. Several different fruit species were planted, but it became apparent that most were not adapted to the climate. As the various species died or were damaged severely, they were replaced with cherry trees. Cherries proved to be quite productive, but nearly the entire orchard was destroyed in the 1940 Armistice Day freeze and was not replanted. Some work also was done with vine crops and, in the early years of the Station, extensive demonstration plantings of garden crops were made. These included, but were not limited to, potatoes, cabbage, tomatoes, beans, and corn.



Tree seedling beds in the State Forest Nursery. Several million trees were distributed from about 1910 through 1952.

After cessation of nursery sales in 1952, only a few trees were produced on the Station. These provided stock to landscape the campus and replace trees that died. In 1964, the Station and the Cooperative Extension Service, Kansas State University, agreed that an Assistant State Forester should have an office at Hays. Mr. Fred Atchison arrived on July 1, 1964 and remained until May 31, 1978. He was replaced by Mr. Jim Strine.

Although not on the Station staff, these men provided valuable services to urban and rural interests.

The Siberian elm (commonly called Chinese elm) tree planted on the Station in 1913 was one of two obtained from China and was the first Siberian elm to be planted in Kansas. This specimen grew rapidly, and the variety gained great popularity. Thousands of seedlings were propagated from this original tree and were distributed throughout the state. Popularity of the species continued until 1931 when a severe blizzard hit western Kansas. Thousands of Siberian elms were injured severely or killed. The next severe damage occurred with the Armistice Day freeze in 1940. Nearly all elms over 16 inches in diameter were killed, and most smaller trees were damaged severely. The old tree that had been planted in 1913 was killed except for one small branch on one side. It was removed the next fall. This puts to rest the rumor that this tree has survived all of the vicissitudes of western Kansas weather and remains alive today. Such is not the case.

STATE PARK

Much of the development of the State Park is intertwined with that of the Station. The legislative act establishing the Station included using part of the land as a public park. The responsibility for development and maintenance of the park fell to the Station (presumably as so directed by the Board of Regents). The first reported action was in 1905, when 43 acres were set aside for park purposes. In that year, the Station graded drives, erected fences and gates, and set out trees that had been growing in the forest nursery. Tree species included redbud, bur oak, black walnut, honey locust, coffee bean, white elm, hackberry, persimmon, green ash, Russian mulberry, Russian olive, red elm, poplar, cottonwood, and soft maple. A grove of Siberian elm was planted west of the original park. Thus today, except for trees immediately adjacent to the creek, most of the trees in Frontier Park can be attributed to the foresight of the early Station administrators. Some of the trees that originally lined the creek, especially those growing west of what is now Main Street, were destroyed as part of the "channel improvement" that took place following the 1951 flood.

In the early 1920's, the Station leased 40 acres of grassland to the City of Hays for the establishment of a golf course. Additional land subsequently was provided so the course could be expanded. Later some of the city recreation areas were established on land east of the wooded park area provided by the Station. These sites were maintained by the city. The Station also granted permission for the city to build the municipal sewage disposal plant on Station-owned land. When that facility was abandoned, the area served as headquarters for the Hays City Park Department. Continuing the previous arrangement, the city paid no fee for using the land.

In 1931, the Kansas Legislature passed House Bill 626 (see Appendix A), which created the "Kansas Frontier Historical Park on the Fort Hays Military Reservation". The new park included the original 43-acre park, the site of the old Fort Hays buildings, and some 120 acres of adjacent Station land. (See Appendix A for a legal description.) Most of this was pasture land that had definite value as part of the experimental work of the Station. The 1931, act also established a Park Board which then took over care and maintenance of the park. But the Station still was involved. In 1933, some 200 men of the Civil Conservation Corps (CCC) began work to improve the park. The Station's Forest Nurseryman was appointed Technical Supervisor of these CCC workers as well as all U.S. Forest Service CCC Camps in Kansas. Additional changes in park management were made in later years when the State Historical Society assumed many of the management responsibilities.

WEATHER RECORDS

Precipitation was recorded at the military reservation beginning in 1868. It is not clear who recorded data after the reservation was abandoned and before the Station was established. However, archived data at the Station show yearly precipitation for the past 133 years (1868 through 2000). Beginning in 1901, observations were taken by Station personnel. When the U.S. Department of Agriculture Dryland Agriculture project was established in 1907, more detailed weather observations became an integral part of that project. Starting in 1910 and, with a lapse of only a few years, continuing until that project closed in 1959, the USDA scientist in charge was responsible for collecting and recording weather data. Thus, precipitation, temperature, wind velocity, and evaporation records have been maintained at the Hays research center since 1907.

Since 1868 yearly precipitation has averaged 22.7 inches. About 77 percent of this total normally falls during the growing season, April through September. Greatest monthly amounts usually occur in May, June, and July. Winter months, November, December, January, and February each average less than one inch of precipitation. Since records have been logged, annual precipitation amounts have ranged from 43.34 inches in 1951 to 9.21 inches in 1956. Average annual snowfall over 107 years has been 19.6 inches. The highest snowfall amounts are received in February and March, averaging 4.8 and 4.3 inches, respectively. Many weather factors other than precipitation are significant, but moisture or lack thereof is the most important factor in dryland crop production.

Average annual mean temperature for 104 years, 1897 through 2000, is 53.9° F, with a mean maximum of 67.2° F and a mean minimum of 40.5° F. Average January mean temperature is below freezing, and July average is about 80° F. As with precipitation, there are large variations from year to year. Except for five years (1908, 1915, 1958, 1972, and 1992), daily maximum temperatures of 100° F or higher were recorded during each summer. Zero or subzero temperatures were recorded in all but three years (1941, 1987, and 1992). Extreme temperatures recorded were 117° F July 13, 1934, and -25° F December 22 and 23, 1989. Average date of the last killing frost in the spring is April 27 and the first killing frost in the fall, October 14, but the frost-free period has varied from 114 to 203 days.

George Griffith was the first to record weather data at the Station. He was the observer from 1901 through February 1903. Mrs. J.G. Haney, wife of Superintendent Haney, was the recorder from March 1903 through April 1904, and George Holden followed from May 1904 through March 1910. A.L. Hallsted, USDA Soil Scientist, came to the Station in 1909 and began keeping more detailed weather data in April 1910. He apparently relinquished this duty to Elliot Gibbons during part of 1918 and 1919, and to A.H. Kern from November 1919 until September 1924. Hallsted again took over, and, with frequent help from Mamie and Maude Hallsted, continued until his retirement in May 1946. Andrew Erhart was the official observer from June 1946 through May 1948, when Paul Brown replaced him as soils project leader. Brown continued until 1956 when he was transferred by the USDA to Montana. Bill Baxter, Assistant to the Superintendent, maintained the weather records from October 1956 until February 1986. Joe Becker, Grounds Manager, is currently the observer and is responsible for all official equipment as well as KSU automated instruments. Official weather observations have been recorded on a daily basis. However the time of day has varied. Since 1946, temperature, precipitation and other data for the previous 24 hours were recorded at 8 AM. Some of the soil scientists determined temperature and relatively humidity several times during the day.

Today, the research center is an official observation point for the National Weather Service and National Oceanic and Atmospheric Administration. Weather stats are reported by local radio and television stations and printed in area newspapers. A variety of weather data can also be located on the center's website (http://www.oznet.ksu.edu/wkarc/) along with links to other related information.

SUPPORT PERSONNEL

Station Superintendents, Department Heads, and research scientists receive recognition from the public, from their professional colleagues, and from the scientific community throughout the world. Research contributions are the only reason for having a research facility. But it seems entirely fitting to pay tribute to the many workers who labor behind the scenes and enable the research staff to accomplish many of the things that lead to their recognition. Throughout most of the past 100 years, the Station staff has included an Assistant to the Superintendent or Administrative Assistant. Some of these individuals had academic rank; others were selected from available clerical workers. As can be seen in the listing on page 40, seven employees held the position between 1904 and 1934 but only four since then. Little is known about the duties of the first seven. But we do know that George Helder, the Assistant to the Superintendent from 1904 to 1913, was appointed Superintendent in 1913. This is the only recorded promotion from the Assistant position. Lawrence Reed, appointed in 1934, was the first to hold the position for an extended time. Following Reed's resignation in 1949, William Baxter became the assistant. Baxter's academic background was in accounting. He was chiefly responsible for maintaining financial records and served admirably throughout most of Bill Duitsman's and Bill Phillips' tenures as Superintendent. He developed a particular interest in weather records. Because of the public's fascination with weather and his accumulation of weather facts, Baxter often was recognized locally as the face of the Experiment Station.

A second important group, collectively called classified employees, includes secretarial support, accountants, maintenance personnel, farm laborers, research technicians, and all of the other individuals that make a successful operation possible. Although one is on dangerous ground to mention certain individuals and not others, efforts have been made to compile a complete roster of all who have ever been employed at the research center. Names were retrieved from the archived records available and are listed in Appendix C. Employees with 20 or more years of service are mentioned here, but nevertheless they represent only a sample of the many dedicated workers who have served at the Station. All employees should be assured that their talents are appreciated and essential to the success of the Research Center. First, a few general observations. Anyone who has not been in the position of Station Superintendent or Head cannot know the satisfaction of waking after an overnight snowstorm and hearing the sound of the snowplow already clearing roads, so cattle can be fed and other necessary functions carried out. I was never sure how the operator negotiated the snowdrifts to get from his home to the Station, but he was always there. Likewise, if weather interrupted electric power, the auxiliary generator in the greenhouse would soon start and provide enough current to prevent freeze damage to the valuable and often irreplaceable plant material. (But where were you when the black cattle got out in the middle of a dark night and had to be found and penned?)

If individuals are to be named it seems only fitting to begin with E.N. Canady who started his service in 1925. Canady was a shop man and blacksmith par excellence. Without his abilities in the shop, it is doubtful that Mr. Aicher could have successfully designed and

constructed the many mechanical devices that were built during his 31 years as Superintendent. Canady remained active until 1949. Another "old-timer" who must be mentioned is F.H. "Fritz" Brueggeman. He came to the Experiment Station in the late 1920's as a young unmarried farm laborer. He lived for a time in the old hotel and was responsible for caring for many of the work animals on the Station. He was promoted to Farm Foreman in the early 1940's. However, he soon realized that he was not suited for this position and left to work elsewhere. He returned to Station employment in 1949 and served as field help on the Weed Control Project until 1965. Howard ImMasche followed Brueggeman as Farm Foreman and served with competence and good humor until 1972. Surely few, if any, others have done the job so well. John Burkhart started his time at the Station in 1942, when he was hired as the tractor and engine mechanic. Those duties were combined with the general shop functions in 1952, and he remained in that position until 1967. Along with Bill Duitsman and several of the project leaders, he designed and built numerous machines adapted for use on the research projects. During this time Harold Boor, first employed as a farm laborer in 1955, came into the shop as an aide, and following Burkhart,s departure, maintained the shop until 1993. He also was responsible for helping design and build specialized equipment needed on the research projects. The last of the century's longtime shop personnel, Ron Wagner, was hired as a farm laborer in 1976. He began helping Harold Boor in the shop and assumed the duties following Boor's retirement. Trimming trees, hedges, and other ornamentals was Seraphin Walters' responsibility for more than 35 years.

Many outstanding classified employees were involved directly with the research scientists in conducting field and laboratory experiments. Certainly there has never been a more dedicated and competent technician than Ralph Dreiling, who served the Livestock Research Project for 40 plus years. And we cannot ignore the consummate cowboy, Glenn Spicer, who was in charge of the beef cattle cow herd for some 20 years. Jim Leiker began in 1949 and for nearly a half century was a virtual fixture in the greenhouses. Most of his career was as a technician on the Entomology project. He seemed to be always on hand, be it workday, weekend, or holiday.

And so, where to stop. The list of employees hired between 1951 and 1980 and who worked for 20 or more years includes Joe Becker, Robert Cline, John Dome, Dennis Ernst, Robert Fields, Harvey Jansonius, Tom Lang, Marvin Leiker, Jerold Penka, John Rhine, and perhaps others I have missed. Because I used an arbitrary cut-off date of 1980, many hired since then and still giving dedicated service are not identified by name. As pointed out in previous sections, longevity in itself does not assure productivity, but the task is more difficult without continuity. The Station is surely unique in having an outstanding group of employees to support the research staff.



Ralph Dreiling at the feed mill control console.

SOCIAL ACTIVITIES AND COMMUNITY INVOLVEMENT

Certainly some scientists interviewed for staff positions declined to accept the job because they felt they would be isolated professionally at an outlying location. In nearly all cases, if they accepted a position and came to the Station, they soon discovered the excellent research climate. Likewise, their spouses may have felt they would be socially and intellectually isolated. In 1935, Mrs. Edith Aicher, wife of Superintendent L.C. Aicher, organized the Station Wives Club. She hoped to provide a rather formal study club atmosphere where staff wives would find interest outside the home. The emphasis changed with passing years, and the meetings became more simple social gatherings. Interest waned as more wives accepted employment in the community and their time with their families had high priority. By the mid-1970's, the club had few activities.

At a time when most farm wives had little involvement in the active management of the livestock operations, Mr. Aicher initiated a "Women's Program" held at the same time as Roundup. These yearly programs involved entertainment and demonstrations such as bread making, meat selection, and meat cutting. The change was gradual, but by the late 1970's, more women were attending the beef cattle program than the women's activities. and the separate program was discontinued.

Although no formal record exists of community activities and involvement by the Station staff, there is no doubt that nearly all were, and still are, active members of the Hays community. Staff personnel and their families found time to be Boy Scout, Girl Scout, and 4-H Club members and leaders. Little League baseball and other sports activities found ready support. Several staff members were active in their churches and civic clubs such as Rotary, Lions, and Optimists. Good relationships between the Station and the local community were fostered through Chamber of Commerce memberships and Board of Education activities,

CONCLUSIONS

As I neared the end of this writing process an acquaintance posed this question: "What is the most significant contribution of the Station during the past 100 years?" Of course, there is no one answer to that question. I could discuss with him crop variety improvements; crop germplasm lines that would, in turn, lead to more improved varieties and hybrids; beef cattle research that resulted in vastly enhanced production efficiency both in the feedlot and on native grass; studies in soil management, weed control, and many other research endeavors that will improve the lot of Kansas farmers, add to scientific agricultural literature, and through applications of this new knowledge help feed a hungry world. But those statements still do not adequately respond to his question. Perhaps we should merely say that the Agricultural Research Center-Hays has provided research leadership in the past and will continue to do so in the future.

APPENDIX A. LEGISLATIVE AUTHORIZATION AND RESOLUTIONS

In 1889, the Secretary of the Interior directed the suspension of action on the Fort Hays Military Reservation to await the action of Congress in regard to it. In February 1895, Representative Jno. Schlyer of Ellis County introduced a resolution in the House of Representatives of the Kansas Legislature, which was adopted. The Senate concurred on February 8, 1895. The Resolution stated:

> WHEREAS, The experience of the settlers upon the plains of western Kansas, covering a period of more than twenty years, has demonstrated conclusively that agriculture cannot be pursued with profit under existing natural conditions, and that artificial means and methods must be substituted therefore; and
> WHEREAS, The tests and experiments required to determine the fitness of new methods applicable to these higher altitudes and limited rainfall cannot be made at the Agricultural College of the state; and
> WHEREAS, The Fort Hays military reservation, at an altitude of 2000 feet above sea-level, contains a valuable body of native timber that should be preserved to posterity, and the land of said reservation is admirably adapted for such experiments in agriculture as are required in the premises; and
> WHEREAS, The buildings upon said military reservation, formerly used as residences for officers and their families, barracks for troops, storehouses, etc., are large and commodious, but cannot be moved without destruction of their value, but in their position are of great value, and could be used, with little additional repairs, for the purposes of a branch of the State Normal School; and
> WHEREAS, The location of a branch of the State Normal School at this place would be central and convenient for the whole of the north half of the state; and
> WHEREAS, The said military reservation has long since been abandoned by the United States government as a military post; now, therefore, be it
> Resolved, by the house of representatives of the state of Kansas, the senate concurring therein, That our senators and representatives in Congress are hereby requested to secure the passage of an act of Congress donating the said Fort Hays military reservation to the state of Kansas for the following public purposes: (1) For a western branch of the Kansas Agricultural College; (2) for a western branch of the Kansas State Normal Institute; (3) for a public park.
> Resolved, further, That the secretary of state is hereby instructed to transmit a copy of these resolutions to the president of the United States senate, the speaker of the house of representatives, and to each senator and representative in Congress from the state of Kansas.

On Saturday, February 23, 1895, a copy of said concurrent resolution was laid before the Senate of the United States by the Vice President. It was referred to the Committee on Public Lands and accepted by Senate Bill 2799 introduced by Senator Martin; it reads as follows:

> Be it enacted, etc.: That the abandoned Fort Hays military reservation and all the improvements thereon, situated in the state of Kansas, be and the same is hereby granted to said state, upon the conditions that said state shall establish and maintain perpetually thereon, first, a western branch of the Kansas Agricultural College; second, a western branch of the Kansas State Normal Institute, and that in connection therewith the said reservation shall be used and maintained as a public park; provided, that said state shall, within five years from and after the passage of this act, accept this grant and shall by proper legislative action establish on said reservation western branches of the Kansas Agricultural College and the Kansas State Normal Institute; and whenever the lands shall cease to be used by said state for the purposes herein mentioned the same shall revert to the United States.

This bill passed the Senate on February 26 and the House on March 2, 1895. Congress adjourned on March 4 and the bill failed to receive the President's signature. Even though Senate Bill 2799 was not signed, the district land officers were advised by a telegram dated March 22, 1895 that lands were withdrawn from settlement to give opportunity for further legislation.

A bill similar to No. 2799 was introduced in the 54th Congress and reported favorably to the House of Representatives by Mr. Charles Curtis from the Committee on Public Lands. This bill did not become a law, although it again was reported favorably by the second session of the 54th Congress. A similar bill was introduced in the 55th Congress and passed the Senate but did not come up in the House. It was introduced again in the 56th Congress, in the Senate by Sena-

tor Harris and in the House by Congressman Reeder, and became a law on March 28, 1900. As finally passed, it reads as follows:

A BILL granting to the state of Kansas the abandoned Fort Hays military reservation in said state, for the purpose of establishing an experimental station of the Kansas Agricultural college and a western branch of the Kansas State Normal School thereon, and a public park. Be it enacted, That the abandoned Fort Hays reservation, and all improvements thereon, situated in the state of Kansas, be and the same is hereby granted to said state upon the conditions that said state shall establish and maintain perpetually thereon, (1) an experimental station of the Kansas Agricultural College, and (2) a western branch of the Kansas State Normal School, and that, in connection therewith, the said reservation shall be used and maintained as a public park; provided, that said state shall, within five years from and after passage of this act, accept this grant, and shall, by proper legislative action, establish on said reservation an experiment station of the Kansas Agricultural College, and a western branch of the Kansas State Normal School; and when ever the lands shall cease to be used by said state for the purposes herein mentioned, the same shall revert to the United States; provided further, that the provisions of this act shall not apply to any tract or tracts within the limits of said reservation to which valid claims have attached by settlement or otherwise under any public-land laws of the United States.

Acting under this law, the 1901 session of the Kansas Legislature passed the following joint resolution:

SECTION 1. That the state of Kansas hereby accepts from the United States the abandoned Fort Hays military reservation, as provided in an act of Congress relating thereto, approved March 27, 1900.
SECTION 2. That the provisions of the act of Congress, 'An act granting to the state of Kansas the abandoned Fort Hays military reservation, in said state, for the purpose of establishing an experimental station of the Kansas Agricultural College and a western branch of the Kansas State Normal School thereon and a public park,' approved March 27, 1900, are hereby accepted by the state of Kansas.
SECTION 3. That upon the approval of this act by the governor, he is requested to transmit a certified copy of the same to the secretary of the interior of the United States.

Approved February 7, 1901.

Copy transmitted to Secretary of Interior February 7, 1901.

The Legislature of 1901 passed an act in respect to the division of the reservation, making appropriations for the branch station and the branch normal school. This act reads as follows:

Be it enacted by the Legislature of the State of Kansas:

SECTION 1. The board of regents of the State Agricultural College and of the State Normal School, respectively, are hereby authorized to locate and establish an experimental station of the State Agricultural College and a branch or auxiliary of the State Normal School on the Fort Hays military reservation.
SECTION 2. The following described tracts of land lying within the limits of the reservation aforesaid, to wit: Section 36, township 13 south, range 19 west; section 31, township 13 south, range 18 west; section 1, township 14 south, range 19 west; sections 6 and 8, the east half of section 7, the north half of section 17, and the northeast quarter of section 18, all in township 14 south, range 18 west, are hereby placed under the direction of the regents of the State Normal School. It shall be their duty to lease or rent the said lands to the best advantage, and all moneys derived from rents for such lands shall be collected by the regents aforesaid, who shall deposit the same with the treasurer of the board, to be expended by the said board of regents for the equipment and maintenance of said auxiliary of the State Normal School.
SECTION 3. All the remaining lands of the reservation aforesaid are hereby placed under the direction of the board of regents of the State Agricultural College, except the north hall of section 5, town ship 14 south, range 18 west, which with the buildings thereon, shall be used jointly as may be determined by the boards of regents of the institutions aforesaid.
SECTION 4. The said board of regents of the State Normal School shall employ a principal and such assistant teachers and janitors as the needs of the school may demand; shall prescribe the course of study, not extending over more than two years, conditions of admission, and such other regulations as may be required for its successful conduct; provided, that such course of study shall embrace only such branches as may prepare pupils for the advanced academic and professional work provided at the State Normal School at Emporia.

SECTION 5. All persons meeting the requirements for admission prescribed by the board of regents shall be admitted to said school; and on declaring their intention to fit themselves to teach in the schools of Kansas shall be exempt from all fees, save a small matriculation fee, which the board of regents may require. Students not intending to teach may be charged a reasonable fee, at the discretion of the board.
SECTION 6. Any person of good moral character over sixteen years of age, having been in actual attendance at least twenty weeks at the above-named school, and having completed the course of study prescribed by the said board of regents, shall be awarded a certificate which shall be a legal certificate to teach in any of the public schools of the state except high schools, and good for one year. Said certificate shall also admit the holder to the third year's work at the State Normal School at Emporia without examination.
SECTION 7. The president of the State Normal School shall be president of said auxiliary normal school, with such duties and responsibilities as the board of regents may determine.
SECTION 8. The sum of $7,000 is hereby appropriated for the fiscal year ending June 30, 1902, and the sum of $5,000 for the fiscal year ending June 30, 1903, is hereby appropriated, for the current expenses and improvements of said auxiliary normal school, the said amounts to be expended under the direction of the board of regents of the State Normal School.
SECTION 9. The board of regents of the State Agricultural College is hereby authorized to locate and establish on the reservation aforesaid an experimental station of the Agricultural College, and shall adopt such measures as may be necessary to place the same in successful operation and to preserve the land upon which the native timber is now growing as a public park.
SECTION 10. To carry out the provisions of section 9 of this act, the sum of $3,000 is hereby appropriated for the fiscal year ending June 30, 1902, and $3,000 for the fiscal year ending June 30, 1903.
SECTION 11. All sums of money payable out of the appropriations specified in section 8 of this act shall be upon vouchers approved by the board of regents of the State Normal School; and all sums payable out of the appropriations specified in Section 10 shall be upon vouchers approved by the board of regents of the State Agricultural College.
SECTION 12. The auditor of state is hereby authorized to draw his warrants on the treasurer of state for the several sums and purposes specified in this act upon verified vouchers approved by the board of regents of the State Normal School or the State Agricultural College; provided, that no portion of the money appropriated in this act shall be expended by the board of regents until the attorney general of the state of Kansas shall first notify the governor and the board of regents that the title to the land in said reservation is unimpaired, and the land is available under the terms of the act of Congress ceding said reservation to the state.
SECTION 13. This act shall take effect and be in force from and after its publication in the official state paper.

Approved February 26, 1901

Published in official state paper. March 1, 1901.

After the Army no longer occupied the reservation, but before the land was given to Kansas, some of the land was filed upon and occupied by settlers. When the state accepted the reservation, these claims constituted a flaw in the title, although deeds to the land had not been granted.

Taking cognizance of this situation, the regents passed the following resolution on April 4, 1901:

Moved by Regent Stewart to adopt the following resolution: Whereas the Legislature at its recent session passed an act accepting from the United States the tract of land known as the Fort Hays Military Reservation and whereas the grant of said land to the state of Kansas provides that it should be utilized by the State Normal School and the Agricultural College and whereas the act of acceptance provides that no money shall be expended on said land until the attorney general shall find that the state can have a good title to all of said land and whereas it has come to be knowledge of this board that a consider able portion of said land is held and claimed by private parties and that the citizens of Hays in connection with the attorney general are now endeavoring to secure the relinquishment of the present claim ants on said land, therefore be it resolved that a committee of four be appointed from the board with direction to call upon the attorney general and if after consultation with him it shall be deemed best shall meet with the regents of the State Normal School at Hays next week and act in concert with them and assist the citizens there in securing the release of the claims on said reservation lands, but under no circumstances shall any arrangements be made whereby funds of this college or the state of Kansas shall be used for securing said releases or for making any improvements on said land until such time as the title of all such

land shall be vested in the state. Moved by Regent Satterthwaite that the committee consist of Regents McDowell, Coburn, Stewart and Fairchild. Amendment carried. Resolution as amended carried.

The matter finally was settled by the Board of Regents executing leases to the claimants to permit them to remain on the land from three to five years in consideration of which all future claim was relinquished.

Plans for the management of the Branch Station were set forth in the following resolution adopted by the Board of Regents on December 13, 1901:

Resolved, That the president of the board of regents shall appoint a regent, who shall, under the direction of the board, have special charge of all matters pertaining to the Fort Hays reservation in behalf of the Agricultural College, the Experiment Station Council to direct all experiments, subject to the approval of the board.
Resolved, That the crop experiments and such other experiments as can be provided for, be begun in the year 1902 on as liberal a scale as circumstances and the funds at our command permit; and that all seeding, cultivation, harvesting, storing, sale and purchase of commodities, or of livestock and its feeding, pertaining to experimental work, and all records in reference thereto, be under the immediate supervision and direction of a competent man, who shall be stationed at Hays so much of the time as may be necessary for best doing the work contemplated.
Resolved, That such repairs be made upon the buildings on the Fort Hays reservation as shall make them available for use, and that a practical farmer be employed, who shall be known as foreman of the farm, and who shall see that all contracts pertaining thereto are fulfilled and all property belonging to the Experiment Station be properly cared for, and shall perform such other duties as shall be assigned to him.
Resolved, That the regent appointed to have charge of the interests of the Experiment Station at Hays shall be paid his per diem and actual and necessary expenses incurred in the performance of such duties, but shall not be allowed mileage.

In 1931, legislation was enacted officially creating the state park and defining its boundaries. The Legislature of 1931 passed House Bill No. 626, which reads in part:

An act relating to and creating the Kansas Frontier Historical Park on the Fort Hays Military Reservation, to be designated and set aside by the State Board of Regents and making appropriation for the preservation and upkeep of same. Be it enacted by the Legislature of the State of Kansas.
SECTION 1. The State Board of Regents shall designate and set aside for park purposes that portion of the Fort Hays Military Reservation which was the site of the buildings of old Fort Hays, together with the remaining buildings of said fort and such other portions of said reservations as the Board may deem to be of particular historical interest of which, in the judgment of said Board, may be favorably situated and well adapted for park purposes, . . . and said Board shall within six months from the effective date of this resolution designate the boundaries of such portions of said reservation so set aside for park purposes.
SECTION 2. The portion of said reservation so set aside and designated by the Board of Regents for park purposes, together with a strip of ground along Big Creek in See. 4, T14, RI8, which has been set aside for park purposes, shall be known as Kansas Frontier Historical Park in the Fort Hays Military Reservation, and said park shall at all times be subject to the general supervision and control of the State Board of Regents, but the active custody and management thereof shall be vested in a Board of Managers, consisting of the chairman of the State Board of Regents, the secretary of the State Historical Society, the President of Kansas State Agricultural College, the President of the Fort Hays Teachers College at Hays, Kansas, and a fifth member appointed by the governor. Said Board of Managers may, subject to revision by the State Board of Regents, make all proper and needful rules and regulations for the use, preservation, improvement, control and maintenance of said park and ground and buildings thereon, and may permit such use of the grounds or any portion thereof, as is not inconsistent with this act or with the purpose for which said grounds have been set aside.

The act also carried an appropriation of $500 for the fiscal year ending June 30, 1932. Over 120 acres were taken for the Frontier Historical Park in addition to the 43 acres previously designated for park purposes. A legal description of the park as established at this time is given in the 1931 Annual Report of the Hays Station. The Park Board was established at that time, and care of the park was transferred from the Station to the new Board.

APPENDIX B. PUBLICATIONS

As pointed out in the Introduction, agricultural research is not complete until the results are communicated to the farming and/or scientific communities. The large number of citations included here indicates considerable success in making results available. Efforts were made to list all of the significant publications, but some may have been omitted inadvertently, and some citations are incomplete. Some of the earliest publications apparently were prepared in-house and do not appear in the publication lists.

Different types of publications serve different but equally important audiences. Scientific journals and books usually include detailed data often collected over several years. Before publishing, most are subjected to review by peer scientists, who also are the major readers. The Kansas Agricultural Experiment Station (KAES), other state agencies (including Cooperative Extension Service), and the U.S. Department of Agriculture (USDA) publish Technical Bulletins, Bulletins, Circulars, Reports of Progress, and others. Over the years, the categories of KAES publications have changed. New ones have been added, others have been eliminated, and the criteria for contents have been modified. Thus, similar reports of research in one area may have been published in two or three categories. The titles of some annual publications also have changed. Most of these emphasize information that is useful to the production agricultural community. Experiment Station scientists also prepare papers for oral presentations at scientific and professional meetings and conferences. These papers often are published in the proceedings of the conferences, but sometimes only abstracts are published in the journal of the sponsoring organization. These are designated (Abst.). In addition, many brief reports and articles are written for publication by the popular press. Most of these are printed in publications not easily referenced so are not included here.

Publications in the following lists are arranged chronologically and then alphabetically by author within each year.

JOURNALS AND BOOKS

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Swanson, A.F. 1941. Relation of Leaf Area to Grain Yield in Sorghum. J. Amer. Soc. Agron. 33:908-914.

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APPENDIX C. ROSTER OF CURRENT AND FORMER EMPLOYEES

Abbott, Phil
Abd El-Hamid, Mosad
Abell, David
Abeln, Leonard
Abernathy, Louis
Ackerman, Jeffrey
Acton, H.S.
Aeilts, Carl
Aeilts, Theodore
Aicher, George
Aicher, Louis C. Jr.
Aikin, S.M.
Akers, Greg
Albertson, Maurice
Albrecht, Gerhard
Alcon, F.L.
Alexander, A.A.
Alexander, E.
Alexander, O.
Allen, F.M.
Allen, Ronald
Allender, Clarence
Almond, Ernest
Almquist, Carl
Alpers, Warren
Alspaugh, Jesse
Anderson, C.O.
Anderson, E.R.
Anderson, Edward
Anderson, Hubert
Anderson, M.
Anderson, Virgil
Andrews, J.L.
Appel, Jon
Applegate, Arthur
Applegate, F.
Applehans, Allen
Arensman, Ben
Arkenberg, B.J.
Armbruster, Betty
Armstrong, Clarence
Armstrong, Ralph
Arnhold, Fred
Arnhold, Gary
Arnhold, LeRoy
Arnold, H.
Arnold, L.N.
Arnold, Sidney
Artman, Peter
Arvilo, Ermilo
Asburn, Pat
Aschwege, Wayne
Ash, B.F.
Ashbaugh, W.L.
Asher, Lloyd
Ashmore, John
Ashmore, Marcia
Ashmore, Silas
Atchison, Fred
Atchison, Fred Jr.
Atkins, John
Atkinson, Irvin
Atteberry, Dean
Augustine, Dennis
Augustine, Michael
Aule, Thomas
Axelson, George
Aye, F.S.
Baird, J.O.
Baird, R.F.
Baker, David
Baker, E.J.
Baker, Ernest
Baker, Herbert
Baker, Sindi
Baker, W.A.
Baldwin, Julie
Ball, Pamela
Banahan, John
Bansfeld, N.
Barber, G.W.
Barber, Roy
Barber, Vernon
Barlow, Richard
Barnard, A.M.
Barnes, Carl
Barnes, Charles
Barnes, John
Barta, Jonathan
Barta, Terry
Bartlett, Lonnie
Bartman, Jerome
Bascom, J.L.
Basgall, Gerald
Batchelder, F.P.
Batman, Erin
Battin, George
Bauer, Merle
Bauer, Steven
Baugham, Eva
Baughman, C.D.
Baughman, Mrs. E.
Baumer, C.
Baumer, Kenneth
Baumer, Ralph
Baus, Raymond
Baxter, A.E.
Baxter, Bill
Baxter, Bird
Baxter, William
Beach, Charles
Bean, William
Beatty, Joseph
Bechard, Mary Beth
Becker, Eldon
Becker, Jacob
Becker, Joseph
Becker, R.B.
Beckum, Bruce
Beem, M.G.
Beeman, J.S.
Beeth, Howard
Befort, Adam
Befort, Alois
Befort, August
Befort, Bill
Befort, Celestine
Befort, Clarence
Befort, Conrad
Befort, Craig
Befort, Edmund
Befort, Ignatius
Befort, John
Befort, Ralph
Befort, Robert
Befort, Tony
Beilman, John
Beilman, P.
Bell, Beulah
Bell, C.W.
Bell, I.
Bell, Leland
Bell, M.J.
Bell, Victor
Bellairs, Harold
Bellingham, Roscoe
Bellis, Nancy
Belman, W.G.
Bender, Curtlin
Bender, J.W.
Bender, William
Bennett, Bill
Bennett, Clifford
Bensfeld, Nick
Berens, Arthur
Berens, Herman
Berens, Jacob
Berens, Otto
Berens, Robert
Berg, James
Bergman, C.W.
Berland, Raymond
Berlin, Guy
Berry, Jerry
Bessant, Clyde
Best, David
Betz, Aaron
Bice, Claude
Bice, Clayton
Bice, Clyde
Biehler, Wayne
Bieker, Alois
Bieker, Aloysius
Bieker, Bernard
Bieker, Eldon
Bieker, Francis
Bieker, Fred
Bieker, Nick
Bieker, Robert
Bieker, Tony
Biery, Jennifer
Biggs, Thomas
Billinger, Mike
Billinger, Peter
Billinger, W.J.
Bills, Frank
Binder, Adolph
Binder, Alois
Binder, Charles
Binder, Donald
Binder, Isidore
Binder, Julius
Binder, Paul
Binder, Robert J.
Binder, Robert R.
Binkley, Royal
Binsfeld, N.
Bird, Cecil
Bird, Eugene
Bird, Forrest
Bird, Ronald
Bishop, Mark
Bissing, Charles
Bissing, Clem
Bissing, Justus
Bissing, Lawrence
Bissing, Pete
Bissing, Raymond
Bissing, Richard
Bissing, Robert
Blackman, William
Blakesley, A.D.
Blakesley, Lee
Blandefield, Paul
Blanding, Terry
Blank, Larry
Blatt, Charles
Blender, Ed
Blender, Francis
Blick, Tillie
Block, Alford
Blocksome, Carolyn
Bobbitt, W.J.
Bode, I.T.
Bodmer, Caleb
Boller, Clyde
Bollig, Frank
Bollig, Marion
Bond, Betty
Boor, Harold
Boos, Al
Boos, Francis
Boos, Henry
Boos, Joe
Boos, John
Boos, Leo
Boos, Peter
Boot, L.
Booth, Robin
Bortic, M.
Boxler, Harold
Boxler, Jerold
Boxler, Martin
Boxler, Mike
Boyd, F.M.
Boyle, Alan
Bozarth, George
Brack, Leon

Bradshaw, John
Brady, Joseph
Brandt, James
Braun, Dan
Braun, David
Braun, Jared
Brenemen, Richard
Brengle, H.I.
Brenn, David
Brenner, David
Brenner, Lance
Brethour, John
Briney, Jerry
Brockman, J.
Brook, Tec
Brooking, O.C.
Brooks, Gail
Brooks, Jess
Brooks, Tina
Brower, George
Brown, Allen
Brown, E.M.
Brown, John
Brown, Leo
Brown, Leroy
Brown, Mark
Brown, Mary
Brown, Ora
Brown, Paul
Brown, Peter
Brown, Ray
Brown, Robert
Brueggeman, F.H.
Brull, Pia
Brull, T.W.
Brumitt, A.
Brumitt, E.J.
Brumitt, Jacob
Brumitt, John
Brumitt, Willard
Brummer, Todd
Brungardt, Allen
Brungardt, Anselm
Brungardt, Anton
Brungardt, B.P.
Brungardt, Carrie
Brungardt, Dominic
Brungardt, James
Brungardt, John
Brungardt, Kenneth
Brungardt, Paul
Bruns, Roland
Bryan, Dorothy
Bryant, V.C.
Buchanan, Kay
Buchheister, Ione
Buchheister, Jack
Buckles, J.R.
Bugg, William
Bullock, George
Bullock, Linn
Bullock, Milon
Bullock, Roy
Bunch, Richard
Bunker, Eddie
Burbach, Douglas
Burch, John
Burgardt, Clifford
Burgardt, James
Burghardt, Etzel
Burkhart, Beverly
Burkhart, John Jr.
Burkhart, Joyce
Burkhart, M.
Burkhead, L.C.
Burkholder, A.L.
Burkholder, Arthur
Burnett, Don
Burnett, R.I.
Burns, Henry
Burns, Joseph
Burns, Thomas
Burris, Irvin
Burris, Less
Burton, Roy
Burtscher, A.W.
Bush, Lex
Butcher, Antone
Butler, John
Butler, Wesley
Butts, W.D.
Bye, F.S.
Byerley, Thomas
Byers, Hugh
Byers, Joseph
Byers, R.H.
Byron, George
Caceres, Jacobo
Caldwell, Fred
Caldwell, R.E.
Calir, Harold
Callahan, J.P.
Callahan, Ronald
Callahan, William
Calvert, F.W.
Calvert, Will
Campbell, Jack
Campbell, Lloyd
Campbell, Martin
Campbell, O.P.
Campmire, Melvin
Canaday, E.N.
Canaday, Henry
Canaday, Roy
Cane, J.B.
Cane, Reece
Carder, Jeremy
Carlile, T.J.
Carlisle, Clair
Carman, Debra
Carmichael, Gordon
Carnahan, J.R.
Carnahan, Roy
Carpenter, Denis
Carroll, Charles
Carson, Karl
Carter, N.D.
Casady, Alfred
Casady, Robert
Case, Alice
Case, Wilfred
Casebeer, Ernest
Casey, Spencer
Cashman, Marjorie
Casper, James
Cathcart, Lester
Chapman, George
Charvat, Van
Chenowith, Harry
Chinn, Tim
Chittenden, H.G.
Chittenden, Thomas
Choitz, Janna
Chrisler, Delmer
Christiansen, W.N.
Clark, Bert
Clark, Clarence
Clark, Delbert
Clark, E.F.
Clark, Henry
Clark, Joseph
Clark, Maude
Clark, Paul
Clark, Richard
Clark, Russell
Clark, Silas
Clark, Stanley
Clark, W.H.
Clause, Earl
Clay, Byron
Clay, Harry
Clay, Henry
Clay, Mrs. T.
Clay, Ray
Clements, M.C.
Cline, Jason
Cline, Rex
Cline, Robert
Cloud, Noah
Cobb, Darrel
Cobb, Don
Cobb, Gareth
Coble, F.S.
Coe, H.K.
Coffey, A.
Cole, Anthony
Cole, E.C.
Cole, Lawrence
Collins, H.S.
Collins, Lewis
Collins, Melissa
Colliver, A.D.
Colson, Peggy
Compton, Swan
Conaway, Thomas
Conger, Duane
Conklin, B.J.
Conklin, George
Connelly, Bill
Connelly, W.E.
Conner, A.
Conner, Ann
Conner, Donald
Connor, Wesley
Conrad, Herbert
Conway, John Jr.
Cook, Kenneth
Coons, J.W.
Cooper, Raymond
Cooper, Roy
Copeland, Paul
Copp, O.D.
Cordill, Nicole
Costner, John
Coulter, Earl
Countrysman, D.
Cowan, John
Cox, George
Coyle, F.S.
Coyne, Patrick
Cragg, F.C.
Craig, J.M.
Craigmile, H.S.
Craine, Rosalie
Cramer, George
Crannell, Max
Cravens, George
Crawford, Donna
Crawford, Keith
Crawford, Linda
Crawford, Lorie
Creamer, Duane
Creighton, Alex
Creighton, Don
Creighton, Keith
Crissman, H.E.
Critchett, A.
Crocker, Benjamin
Crocker, Carl
Crocker, G.
Cron, A.B.
Crosby, Steven
Crosly, F.L.
Crow, J.
Crowl, Clarence
Culley, Kevin
Cummings, A.G.
Cunningham, C.B.
Cunningham, C.C.
Cunningham, F.L.
Currie, Randall
Curry, James
Curry, Leon
Curtis, John
Curtis, LeRoy
Curtis, Phillip
Curtis, William
Cushman, Frank
Custer, Ray
Cutler, Carrie
Dallna, Leo
Dana, A.
Danielson, Albert
Darland, Raymond
Datson, Will
Daugherty, Delwin
Daugherty, John
Davenport, C.A.
David, Leonard
Davidson, A.P.

Davidson, Vester
Davis, Ed
Davis, J.C.
Davis, John
Davis, Lavern
Davis, Mervin
Dawson, J.H.
Dazey, A.C.
Dazey, Ray
Deane, Robert
DeBoer, Guy
Dechant, A.
Dechant, Alvin
Dechant, E.
Dechant, Hayes
Dechant, Herbert
Dechant, John
Dechant, Nicholas
Dechant, Peter
Dechant, Tony
Dechant, Walter
Dechant, Wayne
Deeds, William
DeForest, Elbert
DeForest, Kenneth
Deines, Chad
Deisher, Douglas
Dellett, Fred
Dempsey, Michael
Denia, Elgin
Denning, Gene
Denning, Gilbert
Denning, Melvin
Denning, Nick
Denning, Robert
Denning, Roger
DeWees, Bergen
Deyoe, Carrol
Dible, Diana
Dick, Alex
Dick, S.W.
Dickerson, Charles
Diehl, T.M.
Dies, Barry
DiLiberto, V.
Dilley, Steven
Dinges, Alex
Dinges, Alfred
Dinges, Charles
Dinges, Edward
Dinkel, Aegidius
Dinkel, Brent
Dinkel, Frank
Dinkel, Gerard
Dinkel, Henry
Dinkel, Joseph
Dinkel, Markus
Dinkel, Steve
Dister, Douglas
Dixon, W.C.
Dodd, O.M.
Dodge, C.H.
Dodge, O.D.
Dodge, Walter
Dodrill, Ellsworth
Doerfler, Alex
Doerfler, C.J.
Doerfler, James
Doerfler, Joe
Doerfler, Marvin
Dome, John Jr.
Domey, Charles
Domey, John
Domey, William
Donaldson, L.
Dorney, John
Dorney, Wilfred
Dorney, William
Dorsch, Thomas
Doty, J.O.
Dougherty, Elmer
Downing, Robert
Downs, Arthur
Downs, Walter
Doyle, Joe
Drake, Arlyn
Drake, Herbert
Drake, R.R.
Drees, Mildred
Drees, Thomas
Dreher, M.A.
Dreher, Melvin
Dreher, Philip
Dreher, Victor
Dreiling, Ben
Dreiling, Chris
Dreiling, David
Dreiling, Fred
Dreiling, Gilbert
Dreiling, Joe
Dreiling, Larry
Dreiling, Leslie
Dreiling, Louis
Dreiling, M.P.
Dreiling, Neil
Dreiling, Ralph
Dreiling, Rudolph
Dreiling, William
Driskell, S.R.
Drugan, G.J.
Dubbs, Owen
Dubin, Bob
Duerksen, D.E.
Duesing, Phillip
Duitsman, Wilbert
Dunavan, Clayton
Dunn, George
Durboraw, Glen
Durfee, Dale
Dusin, Karl
Dutton, J.A.
Dye, Dennis
Dyer, Lou
Earley, Donald
Easley, Claude
Easley, S.C.
Eberhart, Ira
Eberle, Marvin
Eberly, Will
Eccleston, Sheldon
Eddleman, Gerald
Ediger, Orlin
Edwards, Charles
Edwards, Dick
Edwards, E.
Ehrlick, Dave
Eickbush, Andrew
Eickbush, Victor
Eickhoff, Nathan
Eilert, John
Eining, Danny
Einsel, Rodney
Ekey, Fauntleroy
Ekey, Sammuel
Elam, Dan
Elder, Harold
Ellenbecker, Frederick
Ellenz, Jana
Elling, O.H.
Ellis, Frederick
Ely, Donald
Ely, W.B.
Embry, Dick
Emery, D.J.
Emery, Russell
Emery, W.E.
Emrich, Frank
Engel, Bruce
Engel, Charles
Engel, Eric
Engle, Alex
Englesby, E.T.
Enlow, Charles
Ensign, Ernest
Ensign, Lester
Epstein, Estelle
Erhart, Andrew
Erickson, C.T.
Erickson, Rex
Ernst, Dennis
Erwin, Joseph
Eshbaugh, D.E.
Eshbaugh, F.P.
Esterdahl, Albert
Eubank, Paul
Everhart, Marion
Everhart, O.
Evers, Carlos
Fabrick, Paul
Fahey, Pat
Fair, Fred
Fairchild, F.P.
Farber, Alan
Farber, Ford
Farlow, J.A.
Feezor, George
Feiden, Jack
Feitz, Joe
Fellers, Harvey
Felten, Pete Jr.
Feltes, Arthur
Feltes, Ed
Feltis, Harold
Feltis, Jim
Feltis, N.J.
Felton, Peter
Fench, Jack
Fenoughty, Frank
Ferguson, Bennett
Fernandez, John
Ficken, Clinton
Ficken, Dale
Fields, Robert
Finch, Douglas
Fink, Ralph
Finkbiner, J.B.
Fischer, Ed
Fischer, Jesse
Fisher, J.P.
Fisher, Peter
Fisher, Ralph
Fisher, Thomas
Fitting, Hans
Fitzgerald, Pat
Fitzpatrick, H.J.
Flanagan, O.K.
Flanders, A.A.
Flanders, J.E.
Flax, L.W.
Fleming, Roy
Floersch, Mathias
Flora, Lowell
Floyd, John
Folkers, Lois
Foltz, C.V.
Foos, Matthew
Ford, Mary
Foreman, George
Foresman, C.E.
Fortune, Jerry
Foss, Dewey
Foster, Bill
Foster, N.T.
Foster, Vernon
Fowler, Wayne
Fox, Charles
Frahm, Richard
France, D.C.
Frank, Edward
Frank, J.S.
Franke, Richard
Franklin, D.E.
Franklin, Woodrow
Frantz, Emma
Freeman, C.
Freeman, G.C.
Freeman, Roy
Freeman, Russell
Freeman, W.M.
Frees, John
Frey, Jon
Fritts, Bernal
Fritts, Gale
Fross, Fred
Fuchs, K.S.
Fuhrman, Stephanie
Fukahara, Harry
Fuller, Leo
Furbeck, Fred
Furbeck, G.E.

Fyfe, Fred
Gabel, Edmund
Gabel, Gilbert
Gabel, Harold
Gabel, Joe
Gabel, Leo
Gabel, Vernon
Gable, Donald Jr.
Gallagher, Pete
Galliart, Henry
Galloway, Walter
Garey, Meryl
Garnett, Thomas
Garrett, Lloyd
Garton, Gail
Garvert, Terry
Gaschler, Dennis
Gaschler, Thomas
Gates, Ryerson
Gatschet, Denise
Gatschet, John
Gatschet, Mark
Gatschet, Ramona
Gauss, Henry
Gauss, Michael
Gearhart, O.D.
Geier, Patrick
Geist, Albert
Geist, August
Geist, Brenda
Geist, James
Geist, Thomas
George, William
Germann, James
Gerstner, Elaine
Gerstner, LeRoy
Gerstner, Norman
Getty, R.E.
Getz, Alec
Getz, John
Gibbons, Elliott
Gibbons, George
Giebler, Alois
Giebler, Felix
Giebler, Mathew
Giebler, Patrick
Giebler, Robert
Giebler, Tony
Gilger, James
Gill, Harold
Gill, Shirley
Gilleece, C.H.
Gilleece, J.S.
Gilmore, Jack
Gilstrap, W.H.
Glascock, Tami
Glassman, Anton
Glassman, Edgar
Glassman, Florian
Glassman, Ludwig
Glassman, Ralph
Gleim, Henry
Gleim, John
Gleissner, Frederick
Glick, Cyrus
Glick, J.M.
Glick, J.W.
Glick, N.A.
Gnad, John
Goble, Harold
Goble, M.
Goetz, Adam
Goetz, Alvin
Goetz, Ernest
Goetz, Eugene
Goetz, Herbert
Goetz, J.P.
Goetz, Vernon
Goetz, Wesley
Goff, Carl
Goff, Lew
Goldforb, Jacob
Goller, George
Good, Larry
Goodrich, Charles
Goodrich, D.T.
Goodrick, Garland
Gordon, J.
Gosney, Abe
Gosser, Lou
Gosser, Roy
Gosser, Wallace
Gottschalk, Adolph
Gottschalk, Clinton
Gottschalk, Curtis
Gottschalk, Mark
Gottschalk, Neil
Gottschalk, Richard
Grabbe, Adolph
Grabbe, Clarence
Grabbe, Edward
Grabbe, Paul
Grabbe, Ron
Graber, John
Graft, P.L.
Graft, Thomas
Graham, Cody
Graham, H.
Green, Connie
Greene, R.
Gregory, Max
Griffing, Ward
Griffing, Willis
Griffith, George
Griffith, Howard
Griffith, Walter
Grisham, A.
Groff, Homer
Groff, Joe
Groff, Walter
Gross, Agnes
Gross, C.F.
Gross, Gail
Gross, George
Gross, Harold
Gross, Harry
Gross, Henry
Gross, Jacob
Gross, James
Gross, John
Gross, Kenneth
Gross, Leo
Gross, Martin
Gross, Paul
Gross, R.
Gross, Theodore
Gross, Vernon
Grossardt, B.R.
Grote, Hilbert
Groves, Clyde
Growe, Kenneth
Growe, Marilyn
Gruner, Cecil
Guardstaff, W.H.
Guernsey, W.H.
Gullett, E.J.
Haas, Francis
Haas, Robert
Hackerott, Glenn
Hackerott, Harold
Hackett, William
Haddock, Donald
Hadley, Ralph
Haefner, Pete
Hager, Kim
Haggard, Jared
Haile, Gorden
Hale, Charles
Hale, Effie
Hale, J.T.
Hale, O.P.
Hale, Selman
Haley, Paula
Hall, Barbara
Hall, Charles
Hall, Joseph
Hall, Richard
Hallsted, A.L.
Hallsted, Mamie
Hallsted, Maude
Halperin, Morris
Hamel, Richard
Hamilton, Mrs. S.E.
Hamilton, Neal
Hamilton, S.E.
Hammer, W.E.
Hammerschmidt, Elden
Hammerschmidt, Otto
Hammerschmidt, Richard
Hancock, Paul
Hancock, Pierce
Haney, J.G.
Hankins, C.R.
Hanley, F.M.
Hann, Joe
Hansen, Alois
Hansen, Keith
Hanson, Kathy
Hardin, J.J.
Hargett, Eric
Hargett, Jennifer
Harkness, J.G.
Harman, Steven
Harmon, S.A.
Harmon, William
Harmoney, Keith
Harper, A.E.
Harper, Arvin
Harper, Betty
Harper, Ernest
Harper, Everett
Harper, John
Harries, George
Harris, Charles
Harris, David
Harris, Frank
Harris, George
Harris, Jack
Harris, John
Harris, L.G.
Harris, Pamela
Harris, Richard
Harrison, Jackie
Harrison, Jewell
Harrison, Joe
Harrison, Rich
Harshbarger, Elwood
Harshbarger, G.E.
Hart, Ernest
Hartley, J.
Harvey, Tom
Hase, Cecil
Hashimoto, Y.
Hauser, Alois
Haverman, Alfred
Hawk, E.S.
Hawkes, E.W.
Hawkins, H.
Hawks, Charles
Hayes, Charles
Haynes, Charles
Hayter, B.
Haywood, Clarence
Haywood, Mary
Hazen, L.E.
Hazlett, Donald
Hazlett, Steven
Hazlett, V.C.
Heaton, Richard
Heger, John
Heinze, Charles
Helder, George
Helm, Robert
Hemphill, Marjorie
Hemphill, Summer
Henderson, William
Henry, G.S.
Hepp, Conrad
Hepp, John
Herl, Justin
Herman, Bob
Herman, Clarence
Herman, Edmond
Herman, Herbert
Herman, Louis
Herman, Mike
Heronema, William
Herrman, Gregory
Herrman, J. Eugene
Herrman, LeRoy

Herrman, Melba
Herrman, Raymond
Herron, Robert
Hertel, Phil
Hertlein, Harvey
Hertlein, John
Hertling, Charles
Hervey, H.H.
Hess, George
Hetler, H.
Heyka, Michael
Hickert, Kevin
Hickman, Harold
Hild, Joseph
Hildebrand, William
Hildwein, H.L.
Hill, George
Hill, H.
Hill, Kevin
Hill, Lyman
Hilton, Ralph
Hinde, R.R.
Hinojos, April
Hoagland, Lee
Hodgson, E.H.
Hodson, Will
Hoffman, Allison
Hoffman, Gerald
Hoffman, J.P.
Hoffman, Jack
Hoffman, John
Hoffman, Melvin
Hoffman, Mike
Hoffman, Ralph
Hoffman, Robert
Hoffman, William
Hogan, Lee
Hogsett, Stanley
Hohn, Marcus
Holden, George
Holland, F.D.
Hollenbeck, G.
Holm, Florian
Holste, Keith
Holyak, John
Honer, Thomas
Hood, Stephen
Hood, Wayne
Hooker, George
Hoop, Ray
Hoover, Cleta
Hoover, John
Hoover, Roy
Hopkins, Terry
Hopp, Terry
Hopper, J.
Horchem, Ronald
Horner, Jackie
Hosley, Roy
House, Clarence
Houser, Theodore
Howard, Jason
Howard, Lance
Howe, John
Hoy, Coronda
Hubbard, C.A. Jr.
Huddleston, Howard
Hughes, Anita
Hughes, E.C.
Huitt, Bill
Huitt, George
Hulstine, Leland
Hunt, G.W.
Hunt, Wayne
Hunter, R. Duane
Hunter, R.A.
Hunzeker, Carl
Hurd, Joseph
Hurley, T.D.
Huser, Bryan
Huslig, James
Huslig, Marcus
Huston, John
Huston, Paul
Hutchison, Herbert Jr.
Hutchison, Patsy
Hwit, Albert
Iliff, Steve
Imel, Charles
ImMasche, Donald
ImMasche, Dorothy
ImMasche, Howard
Immenschuch, A.E.
Indiek, Joni
Ingram, Frank
Irvin, James
Irwin, Devern
Ivil, M.C.
Ivory, Ed
Jaccard, C.R.
Jaccard, J.C.
Jackson, C.F.
Jackson, H.
Jackson, Norman
Jackson, Wells
Jacobini, Horace
Jacobs, Joe
Jacobs, Norman
Jacobs, Ralph
Jacobs, Rita
Jacobs, Robert
Jacobson, Luther
Jaeger, Donagene
Jaeger, H. Rolland
Jaeger, John
Jansonius, Harvey
Jansonius, Troy
Jantzen, Ed
Jarrett, H.J.
Jenkins, Thomas
Jensen, Christian
Jensen, David
Jepsen, Raymond
Jerrett, J.A.
Jewell, A.N.
Jewell, Elwood
Jewell, Frank
Johnson, Adolph
Johnson, David
Johnson, E.L.
Johnson, E.W.
Johnson, Elbert
Johnson, Emil
Johnson, Ernest
Johnson, G.W.
Johnson, George
Johnson, John
Johnson, Norman
Johnson, Oscar
Johnson, Peter
Johnson, Sandra
Johnson, Wayne
Johnston, Charles
Johnston, Milford
Johnston, Walter
Jones, Cliff
Jones, Elgie
Jones, Herbert
Jordan, John
Joy, Rolla
Juenemann, David
Junk, Raymond
Kane, Edna
Karlin, Al
Karlin, Leroy
Karlin, Marvin
Karlin, Norbert
Karlin, Peter
Karlin, Ronald
Karnes, Ray
Karper, R.E.
Karr, David
Kats, Sean
Kays, Fred
Kays, Paul
Kear, Paul
Keberlein, Alvin
Keck, Jimmie
Keefer, William
Keenan, Bob
Keenan, James
Keller, Frank
Keller, Sharon
Keller, Willis
Kellerman, Mark
Kellogg, Doyle
Kelsey, E.B.
Kennedy, Walter
Kent, Clarence
Kent, G. Raymond
Kent, G.C.
Kent, Harry
Kenter, Maurice
Kern, Beth
Kerns, A.H.
Kerr, Gregory
Kesling, Jesse
Kessler, Frank
Khaleeq, Babrak
Khlopeck, Phil
Kiene, F.A.
Kilman, Robert
Kimble, Kirk
Kincaid, Jeffrey
Kinderknecht, Julie
Kinderknecht, Nick
Kinderknecht, Norman Jr.
Kinderknecht, Pamela
Kinderknecht, T.
King, A.A.
King, Bert
King, C.L.
King, Elvin
King, E.R.
King, Leland
King, Max
King, W.A.
Kingsley, H.M.
Kircher, Tonya
Kirk, Kent
Kirk, Larry
Kirkbride, Eddie
Kirkman, M.G.
Kiser, L.
Kiser, Will
Kisner, Steve Jr.
Kitchen, Robert
Klaus, Anton
Klaus, Benedict
Klaus, Cletus
Klaus, Edgar
Klaus, Joe
Klaus, Peter
Klaus, Roy
Klaus, Victor
Klaus, Walter
Klaus, Wilfred
Klaus, William
Klotz, Robert
Knoll, Alex
Knoll, Devon
Knoll, Omer
Knoll, Robert
Knott, Guy
Kobler, Dean
Koerner, Aloysius
Koerner, August
Koerner, Herbert
Koerner, Hilary
Koerner, Leroy
Koerner, Marcell
Kofoid, Eric
Kofoid, Kenneth
Kohl, H.A.
Kohl, Hugo
Kohler, Peter
Kolarik, Stephen
Korbe, Allen
Korbe, Andrew
Korbe, Felix Jr.
Korbe, J.D.
Korbe, Killian
Korbe, Walter
Kraft, Amanda
Kraft, Bill
Kraft, Tammy
Kramer, James
Kramer, John
Kraus, Weston
Krause, Paul

Krause, Stanley
Krauss, Edmond
Krauss, Herman
Krenzin, Myron
Krizek, Tony
Krof, Joseph
Krupp, Joseph
Kuhar, M.E.
Kuhn, Bonnie
Kuhn, Carle
Kuhn, Edmund
Kuhn, Rudy
Kullberg, Mary
Kuo, S.J.
Kurt, Dittman
Kurtz, Dale
Lambert, Clarence
Lambert, Dennis
Lambrecht, Earl
Lambrecht, Irvin
Lamy, William
Lande, H.H.
Landry, A.
Landry, E.
Lane, F.J.
Lane, J.M.
Lang, Greg
Lang, Thomas
Lang, Verlin
Lange, Deborah
Lantow, Dustin
Lappin, Richard
Larson, Ronald
Latas, Timothy
Lauber, Peter
Lauber, Wilfred
Laubhan, Larry
Laughlin, F.
Launchbaugh, John Jr.
Lavitt, Harry
Law, Eugene
Lawler, John
Lawrence, Randall
Lawton, W.W.
Layne, B.A.
Leach, Lee
Leach, William
Leaf, Stanley
Leas, Grant
Leatherman, John
Ledford, James
Lee, Jerry
Leeper, Harold
Legleiter, Gilbert
Legleiter, Julius
Legleiter, Paul
Lehman, Nancy
Leikam, Lucille
Leikam, William
Leiker, Aegidius
Leiker, Agnesine
Leiker, Alfred J.
Leiker, Alfred P.
Leiker, Alois
Leiker, Alois J.
Leiker, Alois S.
Leiker, Alvin
Leiker, Alvin J.
Leiker, Anthony
Leiker, Carl
Leiker, Clarence
Leiker, Clifford
Leiker, Donald
Leiker, Edward
Leiker, Egidius
Leiker, Emil
Leiker, F. John
Leiker, Francis
Leiker, Frank
Leiker, Gilbert
Leiker, Glen
Leiker, Henry
Leiker, Herbert
Leiker, Isidore
Leiker, J.A.
Leiker, James
Leiker, John
Leiker, Karen
Leiker, Killian
Leiker, Lawrence
Leiker, Leo
Leiker, Linus
Leiker, Marvin
Leiker, Melvin
Leiker, Norbert
Leiker, Peter
Leiker, Sylvester
Leiker, Victor
Leiker, Virgil
Leitner, Larry
Lenius, Albert
Leonard, Ralph
Leonhard, J.A.
Lewis, I.
Lietz, William
Lieurance, Dale
Light, L.R.
Likes, Patricia
Lillard, Percy
Lilly, Jack
Lindquist, M.O.
Lindsey, Arlo
Lindsey, Gerald
Linenberger, Alex
Linenberger, Bob
Linenberger, Donald
Linenberger, Rene
Linnenberger, Andy
Linnenberger, Nathan
Lippert, Dale
Littlefield, I.
Litzenberger, Leeland
Livers, Geraldine
Livers, Ronald
Livingston, Michael
Lloyd, T.
Loera, Ruben
Logan, Albert
Logan, Ed
Loomis, A.L.
Lord, George
Lorew, Estel
Lovedilsch, Clarence
Lowe, Alvin
Lowe, Duskin
Lowe, Frank
Loyd, Elden
Loyd, M.S.
Ludington, Clyde
Luea, Gloria
Luebs, Ralph
Luetters, Crystal
Luevano, Ralph
Lumpkins, Melanie
Lund, Henry
Lupton, Gail
Luther, A.L.
Lynd, Alden
Lyons, J.M.
Macek, Ramona
Mackel, Ed
Madden, Eddie
Mader, Francis
Maier, Dan
Malin, R.J.
Malone, E.P.
Malone, Ernest
Maloney, J.L.
Mancuso, John
Mann, J.L.
Mann, W.L.
Mannell, Sean
Mansfield, M.
Mantz, Gregory
Mapes, John
Marcotte, Cary
Marcotte, Terry
Mares, Adolph
Maresch, Marvin
Marjer, Johan
Markel, George
Markwell, Calvin
Marlier, David
Marquardt, A.
Marrow, J.F.
Marshall, Kenneth
Marshall, Max
Martin, Denny
Martin, Leland
Martin, T. Joe
Marvin, Louise
Maska, George
Maska, J.R.
Maska, James
Maska, Joel
Mason, David
Mason, Howard
Massey, Kady
Masters, Richard
Mastin, Mary Lou
Mathes, Marilyn
Mathews, George Jr.
Matthew, Harry
Mauck, Kenton
Maupin, Robert
Mavins, Fred
Maxwell, Caleb
Maxwell, Robert
Maxwell, Walter Jr.
Maydew, Robert
Mayfield, J.I.
Mayhew, Sam
McAfee, Remi
McAleer, Lavern
McAlester, V.D.
McAllister, Donald
McBride, James
McBurney, F.S.
McCabe, LaVonnda
McCall, Dale
McCammon, P.K.
McCane, Charles
McCarthy, Charles
McCarthy, J.J.
McCarthy, Richard
McClelland, C.K.
McClelland, R.C.
McClelland, Thomas
McComas, Carla
McComas, Sharon
McCord, Michael
McCormack, M.J.
McCormick, C.V.
McCourt, George
McCurdy, Gene
McDaneld, Tara
McDonald, Damon
McDonald, E.
McFarlin, Bob
McGowne, Stuart
McGrary, Hugh
McGrath, Arnold
McGrath, Robert
McGuire, William
McInroy, Alfred
McIntosh, Elmer
McIntosh, J.A.
McIntyre, Cheryl
McKanna, Marcus
McKee, Dennis
McKee, Richard
McKinney, Thea
McLain, C.F.
McLain, F.E.
McLain, Sarah
McLatchoy, J.F.
McMahon, Thomas
McNinnery, Fred
McPherson, F.
McReynolds, Harold
McVey, George
McVey, Henry
McVey, James
McVicker, John
McVicker, Robert
McWilliams, Sharell
Mead, Dennis
Mead, Steve
Meade, Doyle
Meade, Earl

Meade, Keith
Meckenstock, Dan
Meder, Edward
Medina, Juan
Meeker, Charles
Meenen, Frederich
Meenen, Lail
Meier, Alex
Meier, Milton
Meis, Thomas
Mellick, Ronald
Mercer, J.W.
Mercer, W.L.
Meredith, Danny
Merkel, L.P.
Merrell, George
Merrill, M.H.
Mestes, William
Meyer, Alvin
Meyer, Charles
Meyer, Frank
Meyer, G.H.
Meyer, H.
Meyer, Joseph
Meyer, P.J.
Meyer, P.K.
Meyers, Mrs. W.F.
Meyers, W.F.
Mick, Carolyn
Mickelson, Charles
Middlehauff, Jack
Middlehoff, C.J.
Mikelson, A.L.
Mikelson, E.M.
Mikelson, F.W.
Milham, J.A.
Milham, Russell
Millard, Floyd
Miller, Agnes
Miller, Alfred
Miller, Alphonse
Miller, Donna
Miller, Ed
Miller, George
Miller, Hal
Miller, Harold
Miller, Jake
Miller, Joe
Miller, John
Miller, Lee
Miller, Marcell
Miller, Robert
Miller, Roger
Miller, Roy
Miller, W.T.
Miller, Will
Milliken, F.B.
Mills, Dean
Mills, G.B.
Mills, H.E.
Mills, Vern
Mindrup, E.F.
Mitchell, W.C.
Mizell, Harold
Moberly, John
Moberly, Joseph
Mock, Allen
Mock, Clarence
Mock, Eddie
Mock, Joseph
Moe, Joseph
Moeder, Englebert
Moffett, Joseph Jr.
Monahan, George
Montague, E.J.
Montgomery, Jerome
Moore, A.B.
Moore, Byrd
Moore, E.W.
Moore, Edmund II
Moore, G.C.
Moore, Hale
Moore, Joyce
Moore, Lily
Moore, Michelle
Moore, Stanley
Moore, Virgil
Moorhous, John Jr.
Moorman, Claude
Morgan, Clem Jr.
Morgan, Clinton
Morphew, Lorin
Morris, George
Morris, William
Morrisey, Billy
Morrison, George
Mort, Ned
Mortensen, A.E.
Morton, L.E.
Morton, Thomas
Moshy, Rolland
Mosier, Ben
Mosier, David
Mosler, William
Moss, Lynn
Motz, Frank
Mowry, Jan
Mudd, Patrick
Mueller, Charles
Mueller, William
Mullen, Charles
Mullen, E.H.
Mumm, Joe
Mundee, Junior
Mundee, Keith
Munsch, Troy
Munsell, Darrell
Munson, F.L.
Munson, Roy
Murphy, Don
Murphy, F.
Murray, Paul
Murray, R.S.
Murrow, F.L.
Myer, Willard
Myers, Bill
Myers, Charles
Myers, Joan
Myers, Johnnie
Mygatt, J.F.
Nash, Leslie
Neal, Reva
Neale, A.S.
Nebel, Warren
Needles, Foster
Needles, Keith
Neeley, Lawrence
Neelly, Rock
Negus, Willis Jr.
Neilsen, John
Nelson, Albert
Nelson, H.E.
Nelson, Joanne
Nelson, John
Nelson, Virgil
Nemecheck, Alvin
Newcomer, Arthur
Newland, John
Nicholas, Henry
Nicholas, Samuel
Nichols, J.L.
Nicholson, Eric
Nicholson, Robert
Nickelson, Marilee
Nickles, W.C.
Nickoloff, K.P.
Nielson, H.T.
Nimmons, John
Nixon, William
Noland, David
Noonan, Mike
Nordyke, Clinton
Norman, E.
Norris, Fred
Norris, Roy
North, Tricia
Northam, Donna
Northam, Francis
Northrup, A.A.
Nulton, Charles
Nulton, John
Nulton, Jonah
Nulton, Rodger
Nulton, Tommy
Nulton, William
Nuzum, Paul
Nyteray, Andrew
O'Brien, M.
O'Gara, Mike
O'Hair, Carl
O'Laughlin, J.
O'Leary, John
Oakley, James
Oakley, William
Ochs, Glenna
Ochs, John
Ochs, Sidney
Okenberry, Gifford
Oldham, T.
Olinger, M.
Olsen, Lee
Olson, Dale
Olson, Kenneth
Orth, Arthur
Orth, Clemens
Orth, Donald
Orth, Frank Jr.
Orth, John Jr.
Oshant, Fred Jr.
Oshant, Jim
Otezza, M.J.
Otott, Roger
Otte, Gary
Ottosson, F.H.
Owens, W.N.
Owensby, James
Page, John
Page, Maynard
Page, Monte
Painter, LaVerne
Palmer, Paul
Parker, E.
Parker, J.B.
Parker, Merle
Parker, Robert
Parks, Ralph
Pate, J.E.
Patterson, A.M.
Patterson, George
Patton, Donald
Patton, Roger
Paul, W.H.
Payne, James
Payne, John
Pearce, H.E.
Pearce, R.L.
Pearson, Lyle
Pearson, M.C.
Pearson, Sylvester
Pederson, John
Pederson, Lester
Pelham, J.L.
Pelzel, Leo
Penka, Jerold
Penny, Lowell
Penny, Skeet
Perkins, Dale
Perkins, F.R.
Perkins, Frank
Perry, E.G.
Peter, Donald
Peters, Adolph
Peters, Fred
Peters, P.
Peterson, Kermit
Peterson, P.
Pfannenstiel, Alfred
Pfannenstiel, Aloys
Pfannenstiel, Bob
Pfannenstiel, Boniface
Pfannenstiel, Casper
Pfannenstiel, Clarence
Pfannenstiel, Earl
Pfannenstiel, Edmund
Pfannenstiel, Edward
Pfannenstiel, Edwin
Pfannenstiel, Frank
Pfannenstiel, Gerald
Pfannenstiel, Gilbert
Pfannenstiel, Harvey

Pfannenstiel, I.P.
Pfannenstiel, Ignatius
Pfannenstiel, Ignatius A.
Pfannenstiel, Irwin
Pfannenstiel, Jake
Pfannenstiel, Joe
Pfannenstiel, John
Pfannenstiel, Joseph
Pfannenstiel, Justin
Pfannenstiel, L.
Pfannenstiel, Marvin
Pfannenstiel, Melvin
Pfannenstiel, Mike
Pfannenstiel, Nick F.
Pfannenstiel, Nick N.
Pfannenstiel, Richard
Pfannenstiel, Roy
Pfannenstiel, Severin
Pfannenstiel, Theodore A.
Pfannenstiel, Theodore L.
Pfannenstiel, Travis
Pfannenstiel, Vernon
Pfannenstiel, Walter
Pfeifer, A.
Pfeifer, Allen
Pfeifer, Alvin
Pfeifer, Anthony
Pfeifer, Anthony Jr.
Pfeifer, Dennis
Pfeifer, Herbert
Pfeifer, John
Pfeifer, Josephine
Pfeifer, Leonard
Pfeifer, Norman
Pfeifer, Ralph
Pfeifer, Randall
Pfeifer, Richard
Pfeifer, Tony
Pfeifer, Tony Jr.
Pflughoft, Ronald
Pfortmiller, Richard
Phelps, A.
Philip, Mary Allyn
Philip, W.D. III
Philips, Allen
Phillips, Alex
Phillips, Doris
Phillips, William
Phillis, E.G.
Pickard, Ronald
Picken, C.A.
Pickering, David
Pierce, E.S.
Pierce, E.W.
Pike, Bill
Pike, Kenneth
Pinkham, Daniel
Pisani, S.
Pitman, C.B.
Pivonka, G.H.
Pixley, H.K.
Planas, Joseph
Plumb, Paul
Plumb, Robert
Podlena, Robert
Poland, Lester
Polley, Joseph
Popp, Daryl
Popp, Glenn
Porter, Anna
Porter, C.E.
Porter, C.W.
Porter, Walter
Potter, Leslie
Pottorff, Michael
Pratt, F.B.
Pratt, Normand
Pratt, S.V.
Pratt, V.L.
Preston, Joe
Preston, V.Y.
Price, J.W.
Price, Mary
Prindiville, George
Pritchard, Kevin
Prizer, R.M.
Prockish, Daniel
Prose, Dale
Pruitt, A.H.
Pugh, A.B.
Puruker, O.R.
Purvis, Roy
Quinn, Jerry
Quint, Alois
Quint, John
Quint, Wayne
Radford, Marcie
Rajewski, John
Ramba, W.A.
Ramey, Ceis
Ramey, W.T.
Randa, Elmer
Randolph, Lawrence
Rao, Aluri
Rathbun, Bruce
Ray, William
Reed, Bert
Reed, C.S.
Reed, Clyde
Reed, Lawrence
Reed, M.H.
Reed, Mac
Reed, Ralph
Reed, Thomas
Reidel, John
Reimer, Greg
Revilla, Luis
Reyner, Guy
Reynolds, Clarence
Reynolds, Lester
Rhine, John
Rich, Jonathan
Richards, Joseph
Richardson, A.
Richardson, Joyce
Richter, W.A.
Riddle, Jimmy
Rider, Eugene
Ridgeway, S.C.
Riedel, Arthur
Riedel, Eddie
Riedel, Elmer
Riedel, Julius
Riedel, Matt
Riedel, Richard
Riegel, David
Riegel, Donald
Riegel, Kenneth
Ring, Charles
Ritter, E.E.
Robben, Raymond
Robbins, Melvin
Robinson, Alex
Robinson, Jamie
Robinson, John
Robinson, Stephen
Robison, Wayne
Rogers, A.E.
Rogers, Earl
Rogers, Pat
Rohleder, Edward
Rohleder, Luke
Rohleder, Richard
Rohleder, V.J.
Rohleder, Valerius
Rohr, Adolph
Rohr, Al
Rohr, Alex
Rohr, Alfred
Rohr, Alois
Rohr, Alvin
Rohr, Anthony
Rohr, Clarence
Rohr, Conrad
Rohr, Daniel
Rohr, Eddie
Rohr, Edna
Rohr, Frank
Rohr, Gilbert
Rohr, Jerry
Rohr, John
Rohr, Joseph
Rohr, Julius
Rohr, L.B.
Rohr, LaVern
Rohr, Lawrence
Rohr, Matt
Rohr, Melvin
Rohr, P.J.
Rohr, Philip
Rohr, Ruth
Rohr, Theabold
Rohr, Tony
Rohr, Wilfred
Rohrer, Samuel
Romme, Dale
Root, L.
Rorabaugh, Roscoe
Rose, G.N.
Rose, Selby
Rose, W.S.
Ross, Joyce
Ross, William
Rostocil, Walter
Roth, Cecil
Roth, Linus
Roth, Louis
Roth, Vic
Rothenberger, Paul
Rothenberger, Vern
Rothgeb, B.E.
Rothgeb, Harold
Rousseau, Angela
Royce, Clare
Royse, David
Rubottom, Darrel
Ruda, Robert
Ruder, Albert
Ruder, Alfred
Ruder, B.M.
Ruder, Brian
Ruder, Charles
Ruder, Duane
Ruder, Eugene
Ruder, Fidelis
Ruder, Gene
Ruder, Ig
Ruder, James
Ruder, Marvin
Ruder, Nick
Ruder, Richard
Ruder, Theabold
Ruder, Vernon
Rumpel, Philip
Rumsey, Jack
Runey, Jake
Rupp, Alex
Rupp, Basil
Rupp, Ben
Rupp, Darrel
Rupp, Hank
Rupp, Joe
Rupp, Leo
Rupp, Paul
Rupp, Ralph
Rupp, Richard
Rush, John
Rusk, Walter
Russell, Dean
Russell, Rick
Russell, Stanley
Rutheford, O.L.
Ruthrauff, R.
Ryan, Wendell
Ryberg, Rena
Sack, Eugene
Sack, Leo
Sack, Tony
Sack, Vera
Salisbury, Glenn
Salisbury, Ray
Sallee, James
Sallee, Robert
Salomon, Raff
Samuelson, Earli
Sander, Alice
Sander, Alvin
Sander, Anton
Sander, J.A.
Sander, Melvin

Sanders, Elmer
Sanders, I.
Sanders, John
Sanders, Linus
Sanders, Peter
Sandlin, Rhonda
Sandr, Ray
Sanstedt, S.A.
Sarasy, Lewis
Satterfield, J.W.
Sauer, Frank
Sauer, Fred
Savage, D.A.
Savage, David
Saylor, Bonnie
Sazama, Todd
Schank, John
Scheck, Dennis
Scheck, Pete
Scheck, Raymond
Schecter, Edward
Schesser, Earl
Scheuerman, J.E.
Schewey, Pett
Schieferecke, Joseph
Schilling, F.J.
Schilling, Troy
Schiselbein, Joe Jr.
Schiselbein, John
Schiselbein, Joseph
Schlegel, Edmond
Schlegel, Elman
Schlegel, Eugene
Schlyer, Virgil
Schmeidler, Douglas
Schmeidler, Neal
Schmeidler, Phil
Schmeidler, Scott
Schmeidler, William
Schmidt, Carl
Schmidt, Curtis
Schmidt, Francis
Schmidt, Gary
Schmidt, Gene
Schmidt, George
Schmidt, Harold
Schmidt, Jeffrey
Schmidt, Joe
Schmidt, John
Schmidt, M.A.
Schmidt, M.G.
Schmidt, Marion
Schmidt, Scott
Schmidt, Stanley
Schmidt, Stephen
Schmidt, Vernon
Schmidt, Vincent
Schmidt, Warren
Schmidt, William
Schmidtberger, Wayne
Schneider, A.
Schneider, C.
Schneweis, Douglas
Schnose, Gregory
Schoendaller, Frank
Schoenfelt, Jake
Schoenthaler, Arnold
Schoonover, P.C.
Schrage, Michael
Schreiber, James
Schreiner, Daniel
Schreiner, Peter
Schroeder, Randy
Schroeder, W.P.
Schuckman, Adolph
Schuette, Robert
Schuetz, Edmund
Schuhs, Forrest
Schuler, Joe
Schull, C.W.
Schulte, Anton
Schulze, Dennis
Schumacher, Adolph
Schumacher, Anthony
Schumacher, Arlyn
Schumacher, Felix
Schumacher, H.H.
Schumacher, Ignatius
Schumacher, Isidor
Schumacher, Julius
Schumacher, Katherine
Schumacher, Laverne
Schumacher, Marvin
Schumacher, Nicholas
Schumacher, Paulinius
Schumacher, Ralph
Schumacher, Richard
Schumacher, Wendelin
Schumaker, Gary
Schuvie, Pete
Schwartz, D.J.
Schwartz, Paul
Scott, Carl
Scott, H.
Scott, Ira Jr.
Scott, J.L.
Scott, Richard
Scott, Robert
Scuturo, A.
Scymanski, Peter
Seaman, Clayton
Searcy, Billy
Sears, A.
Segovia, Romualdo
Seibel, Darrell
Seifers, Dallas
Seiling, Dale
Seiling, W.T.J.
Seitz, Mike
Seitz, Thomas
Sell, M.
Sellers, Ray
Shade, Arlo
Shade, Clide
Shade, O.
Shade, S.E.
Shade, Will
Sham, L.L.
Shank, Katherine
Shaver, Richard
Shellhart, L.B.
Shepherd, A.
Sherbon, Cliff
Sherbon, J.M.
Sherbon, Max
Sherbon, Milford
Sherbon, Ralph
Sherfick, Walter
Sherley, Clyde
Shield, Hugo
Shippy, Darrell
Shirley, Tony
Shockey, M.M.
Showalter, Jesse
Showman, C.E.
Shutts, Joseph
Sidwell, C.A.
Siemsen, Dorothy
Sigle, Paul
Simmons, John
Simms, William
Simpson, Gay
Sims, Charles
Sipahi, Nihat
Sites, B.
Sites, R.E.
Sites, Sanford
Skinner, Jerry
Slaughter, Dale
Sloop, Elmer
Smiley, John
Smith, C. Lynn
Smith, C.A.
Smith, C.C.
Smith, C.W.
Smith, Carl
Smith, Claud
Smith, Donald
Smith, E.H.
Smith, Earl
Smith, Elmo
Smith, F.M.
Smith, Guido
Smith, Harriet
Smith, J.R.
Smith, Johnnie
Smith, LaMont
Smith, Ralph
Smith, Richard
Smith, Robert
Smith, S.G.
Smith, Timothy
Sneed, Joe
Snook, Ernest
Snyder, Bradley
Snyder, R.C.
Solomon, Fred
Solomon, Robert
Sommers, Lester
Soukup, Thomas
Spanier, Justin
Speck, Thomas
Speirs, John
Spence, L.G.B.
Spencer, Ira
Spencer, James
Spencer, Rachel
Spicer, Don
Spicer, Glen
Spicker, Don
Spreen, John
Spreen, Louis
Spreier, L. LaVern
Springer, Henry
Springer, L.K.
Springston, Scott
Staab, Bill
Staab, Charles
Staab, Craig
Staab, Kirk
Staab, Patrick
Staab, Richard
Staats, Todd
Stabb, Charles
Stahl, A.L.
Stahlman, Phillip
Stang, Clement
Stanley, LeRoy
Starkey, M.A.
Stealy, V.E.
Stecklein, Alex
Stecklein, Edwin
Stecklein, Joe
Stecklein, N.V.
Stecklein, Nicholas
Stecklein, Ralph
Stecklein, Richard
Stecklein, Steve
Stecklein, Victor
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Steed, Charles
Steele, John
Steele, Larry
Stegman, Ashley
Stegman, Wendlin
Stegmeier, William
Stehwein, Robert
Steimel, Lyle
Stein, John
Stenjord, Ivan
Stephen, Cynthia
Stephenson, A.E.
Sterrett, R.R.
Stevens, C.C.
Stevens, F.M.
Stevenson, Leonard
Stevenson, Lori
Stevenson, Robert
Stewart, Alvin
Stewart, Michael
Stewart, O.C.
Stewart, William
Still, H.
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Stirnkorb, Darlene
Stivers, Eda
Stock, Earl
Stock, Harry
Stock, Leo
Stock, Mila

Stock, Ted
Stoffer, W.C.
Stone, J.C.
Stones, Ardith
Storms, Ralph
Stout, Ralph
Stramel, Eugene
Stramel, Frank
Stramel, Jean
Stramel, Jim
Strawn, Thelma
Streeter, Lyman
Stremel, Amanda
Strine, Duane
Strine, James
Stuart, A.C.
Stucky, Tiffany
Studley, Ann
Stump, Fred
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Suarez, Carlos
Suelter, Stanley
Sullivan, Bert
Sullivan, F.
Sullivan, Wallace
Sullivan, Ward
Sultzer, Clyde
Sumacher, Jacob
Sumner, Larry
Sumpter, P.E.
Swanson, Arthur
Swanson, O.S.
Swartzenberg, Louis
Sweat, Martin
Sweetland, Forrest
Sweetland, Leon Jr.
Sweetland, Paul
Swenson, Meredith
Swords, Garry
Taggart, Travis
Tahl, Harry
Tahl, Henry
Talbott, Wade
Tandy, Mark
Tatro, Francis
Taylor, J.E.
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Taylor, Lionel
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TenEyck, A.M.
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Thomas, Arthur
Thomas, Keith
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Thompson, Carlyle
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Thompson, M.L.
Thompson, Roy
Thorman, H.W.
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Timken, Steven
Timmons, F. Leonard
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Titterington, E.W.
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Toepfer, Victor
Tomanek, Gerald
Tong, Shirley
Tonroy, Nathan
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Tuttle, Robert
Tweedy, R.L.
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Tyler, H.
Uark, B.M.
Ulrey, H.
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Unrein, William
Unrine, Phil
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Unruh, Phil
Unruh, Roger
Urban, Frank
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Urban, Joe L.
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Van Doren, Junior
Van Doren, Loyal
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Vanzant, Eric
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Velharticky, Francis
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Weigle, Ben
Weilert, John
Weimer, Claude
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Wilson, Clyde
Wilson, Gene
Wilson, J.L.
Wilson, James
Wilson, Jerry
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Windholz, Richard
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Windholz, Walter
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Winter, Marlene
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Wire, Theodore
Wise, W.
Witt, C.A.
Witt, Marion
Witt, Robert
Wittam, Loyd
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Wolz, J.H.
Wood, Randy
Wood, W.G.
Woodman, J.
Woodsum, George
Woodsworth, M.C.
Wooster, Lyman
Worley, Allan
Worstell, Vard
Woydziak, Matthew
Wright, Earl
Wright, Edmund
Wright, Eugene
Wright, Richard
Wright, Willis
Wynn, Norman
Wynne, Mac
Wynne, William
Xie, Feng-zhou
Yasmer, Lowell
Young, George
Young, Richard
Young, Will
Younger, Adam
Younger, Michael Jr.
Younger, Warren
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Zerfas, Wilfred
Zierlein, Lynn
Zimmerman, Jayme
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Zimmerman, Virgil
Zimmerman, William
Zinger, Sheldon
Zinszer, Harvey Jr.
Zinszer, Junior
Zongker, Jesse

This list of employees was compiled from current personnel records of the Research Center and archived data as available. Although every effort was made to ensure the accuracy of this list, the possibility of omission still exists and is completely unintentional. Names are printed exactly as they were entered into our records.

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